

My Capstone Proposal

“Parent Pal” Mobile Application

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1. Project Overview

In my capstone project, I want to create a mobile application called “ParentPal” that supports new parents in managing and tracking their baby’s development. The app will allow users to log milestones, record daily activities, and set reminders for important tasks. Some Features that I would like to implement are: Growth Tracker - log and compare height and weight against standard growth charts, Sleep Analytics - records sleep patterns and generates insights to help parents optimize their baby’s sleep routine, Health Journal - Track vaccinations, medications, and symptoms to ensure the baby stays healthy, Customizable Notifications - Tailor reminders for activities, nap schedules, and development exercises, Photo & Video Storage - Capture and tag memories directly within the app, linking them to specific milestones, Shared Accounts - Collaborate with a partner, babysitter, or family to manage tasks and share updates, Data Export - Allows users to generate reports for doctors or personal records.

2. Scope

- Have a mobile first, responsive design targeting mobile and desktop
- Protect against SQL injection attacks
- Have authentication and authorization systems
- Consistently use techniques to ensure data integrity and validation
- Perform error handling and logging without exposing internal information to the user
- Use relational, SQL based databases, no ORM or other framework that abstracts SQL (Hibernate, Django, etc) allowed.
- Secure data storage for player account information and any personal data input

Who is the primary intended user of your system? What is the main reason they want to use your system?	
The main users of my system are new and expecting parents. The reason they would use my system is because they want to track their baby's growth, health, and development	
List similar web apps with URLs that you can use for inspiration when building your app. Note the elements you find most similar. You may not simply reproduce an existing site; you must have a unique take on it.	
- While I couldn't find any web apps for managing and tracking a baby's development, I was able to find two links to mobile apps from the app store/google play store. https://apps.apple.com/us/app/baby-tracker/id1444238371 https://sprout-apps.com/apps/baby-tracker/ - These apps are very similar and have the same features, however I don't see many apps that are both a mobile app and a web app 2` `	
What is the primary data entity in your project? (I.e. recipes, game items, surveys, flash cards, budget items, health metrics, etc.)	
The primary data entity in my project is health metrics. Health metrics revolving around a baby's development will be the focus.	
What is the primary functionality that the user will be doing with this data entity? (hint: if you use the word "and" you might not be down to a "primary" function)	
The user will use the health metrics to monitor and analyze their baby's development	
Pick two extended functionalities, different from your core functionality. Briefly outline how it will integrate into your project:	
Discussion forums, messaging between user roles, either item ratings or item favorites, CRUD based event manager/reminder, appointment booking, shopping cart that allows multiple items, increase/decrease quantities and payment processing (e.g. Stripe), AI or ML tool, usage/order history/selling history	
- Messaging System - Enable communication between parents and caregivers	- Event Manager - Schedule and track doctor appointments, vaccinations, and reminders

Specify at least 2 user roles not including “logged out/public”. Include different, non-trivial functionality for each role (i.e. admin can perform a report on all the products on the site, but a regular user can only see their product’s history). Each role must be able to edit its own profile information and preferences

- Regular User (Parent)

- Able to add and track baby milestones
- Able to set reminders and notifications
- Able to manage shared access with caregivers

- Babysitter

- Able to view baby milestones and activities
- Able to log relevant observations
- Able to receive assigned tasks and reminders

Identify the key fields from your core entity and suggest appropriate filtering techniques (more than 1 technique must be used: e.g. substring match, numeric range, etc)

Some of my key fields from my primary entity (health metrics) are:

- The date the data was inputted and updated (date range)
- Height and Weight Metrics of the baby growth (drop down menu)
- Sleep Analysis – How many hours of sleep the baby got (drop down menu)
- Allergies the baby has (drop down with common allergies & text input for allergies not on the common list)
- Medications/Vaccinations the baby has had (text input)
- Images/Videos of baby’s development (import/export from user’s camera roll)
- Camera access to take images/videos (camera button – asks user for permission)
- Observation notes (flag level - Very Important, Important, Normal)
- Data export – for doctors/physicians (Well formatted PDF document)

All web projects must implement protection against session hijacking and XSS.

All web projects must implement accessibility features like alt text, keyboard navigation, colour contrast, form labels, heading structure, ARIA roles or HTML5 landmarks, meaningful text descriptions on links, and ARIA roles and labels on all custom controls. (Test with tools like WAVE, axe DevTools, Pa11y, Lighthouse, etc)

All web projects must allow searching on any of the details (fields) of the core entity, with ranges where applicable. Filters must be provided, and multiple filters must be available simultaneously. The returned search must be displayed in a structured way (grid, table, etc.) that includes pagination of some sort (infinite scroll is allowed, but must be demonstrably triggered during navigation, not pre-fetched and cached).

3. Functionality Chart

Feature	Notes	Parent	Babysitter
Authentication System (Sign In/Sign Out)	- Local user email/password validation - Captcha for failed attempts - Session timeout settings		
Authentication System (Sign Up)	- Validate email address (Confirmation link) - Password Strength Checker - User roll choice (Parent or Babysitter)		
Authentication System (Forgot Password)	- Password reset email (Confirmation link) - Password Strength Checker		
Dashboard	- View Growth Tracker, Sleep Analytics, Health Journal, Feeding, Observation Notes and Assigned Tasks		 (Limited)
Profile Settings (Changes)	- Upload/Change profile picture - Change password - Change email (Revalidate)		
Profile Settings (Deletion)	- Delete Account		
Growth Tracker (Logging)	- Log Weight (lbs or kg's) - Log height (ft or in) - Growth Chart (Based on data entered) - Edit and delete height and weight entries		
Growth Tracker (View)	- View height and weight entries - Filter (By height or weight)		

Growth Tracker (Comparing)	- Compare with Standard Growth Charts (In app charts from reputable sources) - Filter (By day, month or year)		
Sleep Analytics (Logging)	- Log how many hours slept - Log the time baby fell asleep around		
Sleep Analytics (Edit/Delete)	- Edit and delete hours slept and or time they fell asleep around		
Sleep Analytics (Insights)	- Compare results by average hours slept and at which times - Filter (By day, month or year and by lowest hours, highest hours, time of day)		
Health Journal (Sickness/Medications)	- Log medications baby takes or has taken (Tylenol, Advil, Tumms, etc) - Log description of sickness		
Health Journal (Allergies/Vaccinations)	- Log vaccinations - Log allergies		
Feeding Notes (Logging)	- Log Feeding time of day - Log how much baby was fed - Log what was fed - Notes for extra information		
Feeding Notes (Editing/Deleting)	- Edit/Delete feeding logs		
Feeding Notes (Viewing)	- View feeding logs		
Photo/Video (Uploads)	- Upload photos/videos from phone/computer camera roll to milestones		

Photo/Video (Deletion)	- Delete any photo/video from milestones		
Camera Access	- Allow user to open camera app on phone to take photos/videos		
Custom Notifications (Setting)	- Set up sleep reminders (time of day) - Set up feeding reminders (time of day) - Set up doctor's appointment reminders (day and time) - Set up playdates/development game reminders (day and time)		
Custom Notifications (Showing)	- Show notification and which category it belongs to (feeding, sleep time, doctor's Appointment, playdate/development game)		
Shared Accounts	- Share account information with another parent (users partner) - Share account information with a babysitter/caretaker (select information – items with checkmarks)		
Observation Notes (Creating)	- Create observation notes with setting a priority flag to each note (high, medium or low priority)		
Observation Notes (Viewing)	- View observation notes - Filter (high to low priority, date)		
Observation Notes (Edit/Delete)	- Edit/Delete observation notes		

Data Export	- Export all important data to a nicely formatted PDF file		
Assign Tasks/Reminders - Create	- Assign tasks to Babysitter - Set reminders for Babysitter		
Assign Tasks/Reminders - Read	- Read and acknowledge tasks and reminders		

4. Screen Mockups

4.1 Log In w/ Face ID (Mobile) - First screen the user will see



Welcome back!

Email

Password

[Forgot Password?](#)



Face ID

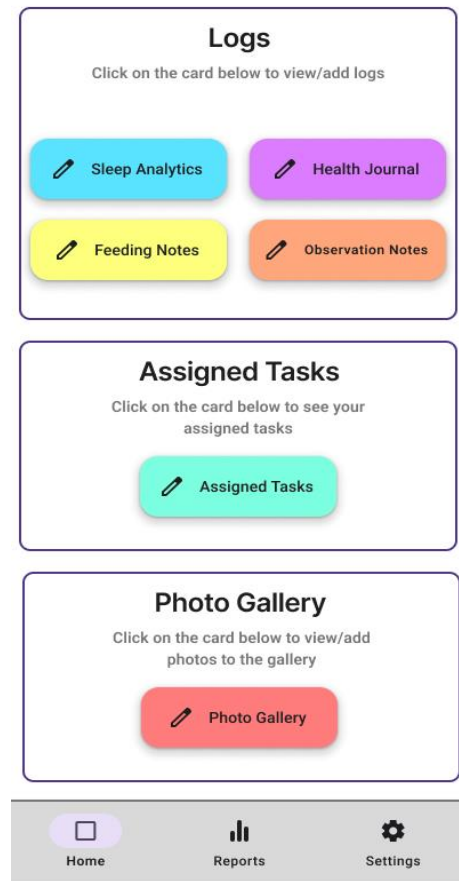
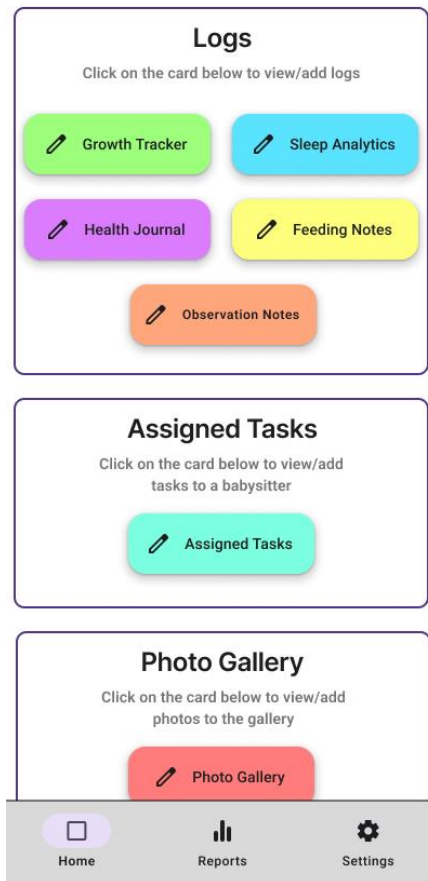
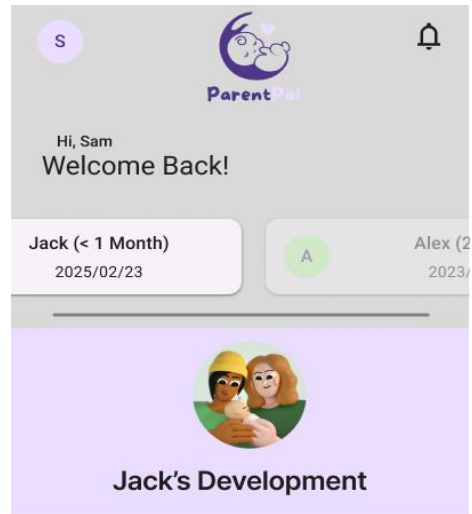
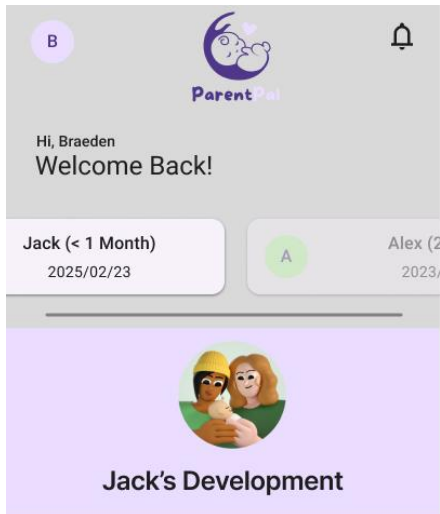
Log In

Are you new here? [Create account](#)

4.2 User Dashboard - Screen after Log In

Parent

Babysitter



4.3 Sign Up - User clicks on create account from bottom of Sign In page

- First three pages to Sign Up are the same for Parent and Babysitter, but then have their own sequence

Add personal info



Create an account

Full Name

Email

Password

Create an account

Already have an account? [Log in](#)

Enter 4-digit verification code

←

Enter the 4-digit code

Your passcode has been sent to your email address
braeden.lyman1@mohawkcollege.ca

0

Resent OTP will be available in 00:12 sec

From Messages
050518

1	2 ABC	3 DEF
4 GHI	5 JKL	6 MNO
7 PQRS	8 TUV	9 WXYZ
	0	<X

Successfully verified



Successfully verified

Let's start creating your account so you can start using the app

Start

4.3.1 Sign Up - Parent Sequence

User selects parent

←

What brings you to our app?

P I am a parent
I am here to keep track and manage my baby's growth and development

B I am a babysitter
I am here to babysit and stay on track for the baby's schedule and development

Next

More info about parent

←

Tell us more about yourself



Date of Birth
1997/01/04

Gender
Male

Next

Adding their child

←

Add your baby

B Baby (0-12 months)

T Toddler (1-3 years)

P Preschool (3-5 years)

G Gradeschool (5-12 years)

Next

More info about child

←

Tell us more about your baby



Name
Jack

Date of Birth
2025/02/23

Gender
Male

Finish

Profile completed

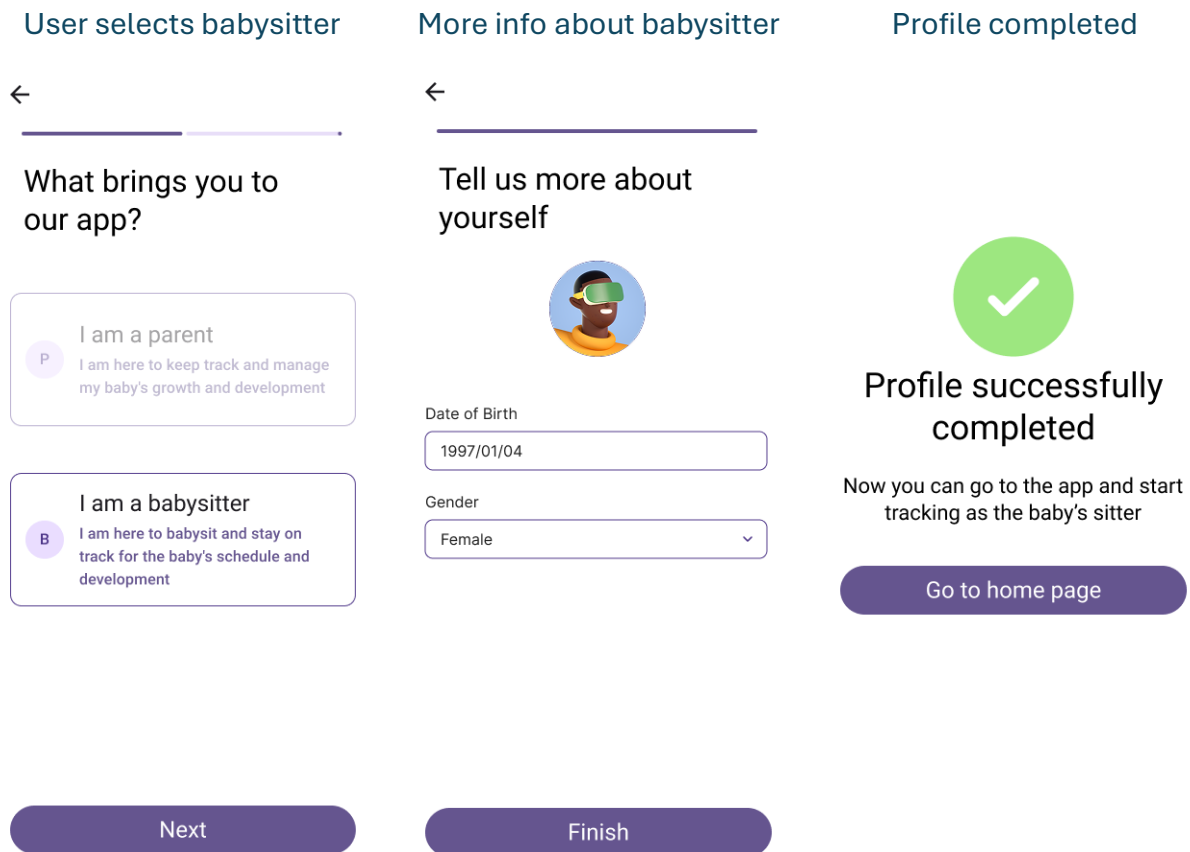


Profile successfully completed

Now you can go to the app and start tracking your baby's development

Go to home page

4.3.2 Sign Up - Babysitter Sequence



4.4 Forget Password - Sends user a link for if they forgot their password

User enters email

←

Recover password

Enter your account email to recover password

Email

Send restore link

Enters 4 digits sent to email

←

Enter the 4-digit code

Your passcode has been sent to your email address
braeden.lyman1@mohawkcollege.ca

0

Resent OTP will be available in 00:12 sec

From Messages
050518

1 ABC	2 DEF	3 GHI
4 JKL	5 MNO	6 PQRS
7 TUV	8 WXYZ	9 0

0

Enters new password

←

Enter new password

Password

✓ 8 characters length
✓ 1 number
✓ 1 uppercase letter

Repeat Password

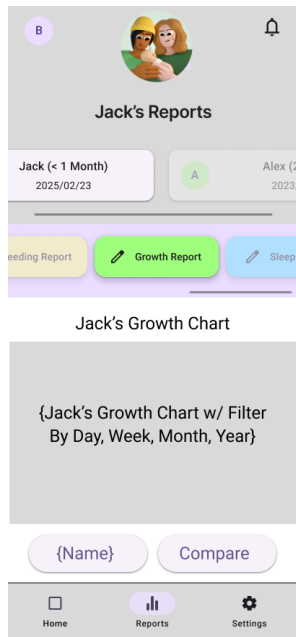
Confirm

4.5 Reports - Shows reports on child's development.

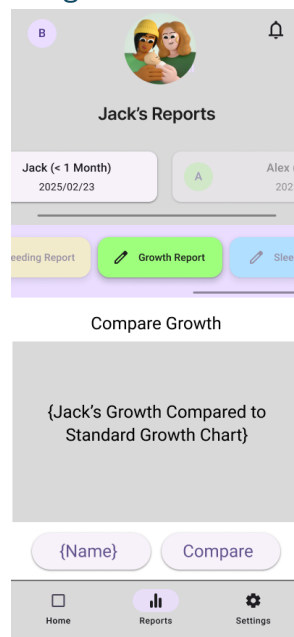
- Parent has growth, sleep, health and feeding reports

- Babysitter has sleep, health and feeding reports

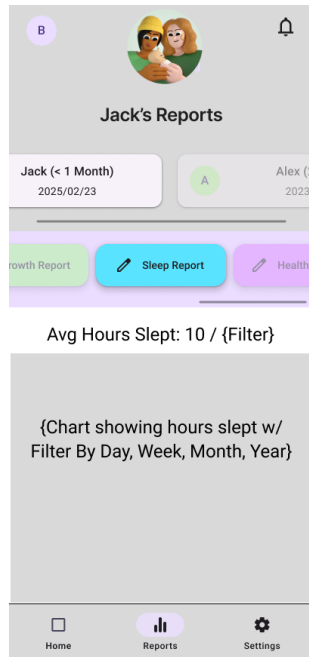
Child's growth chart



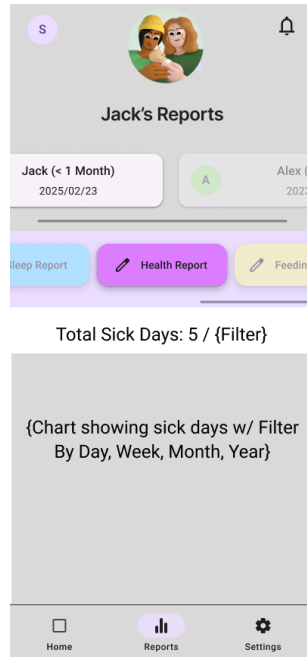
Comparing to standard growth charts



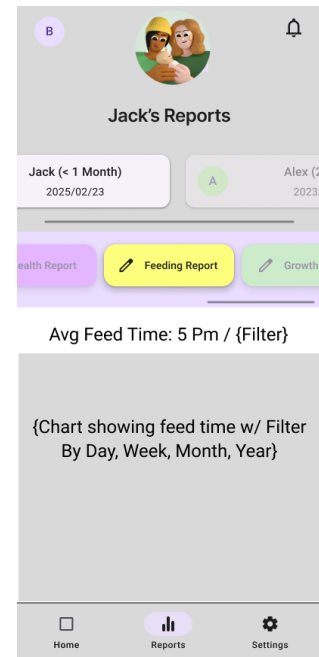
Avg hours slept



Total sick days count



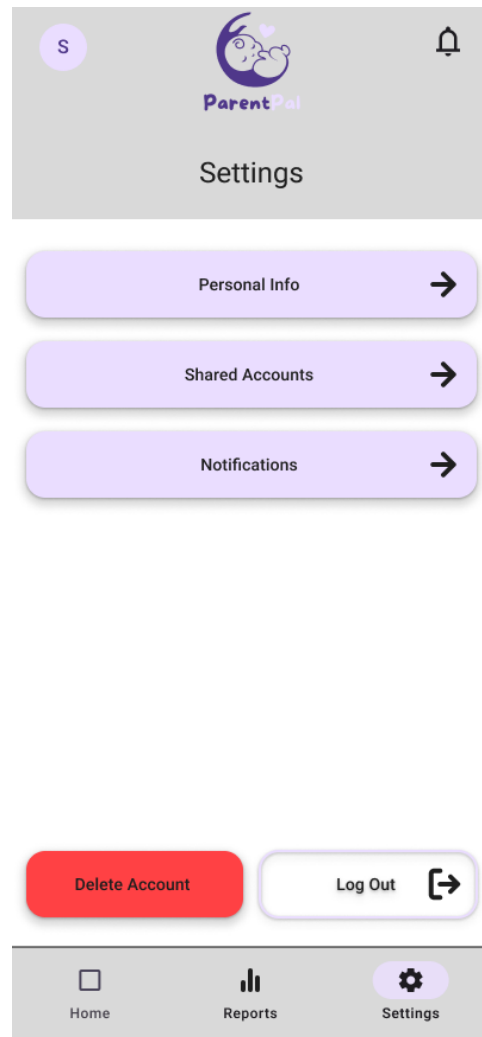
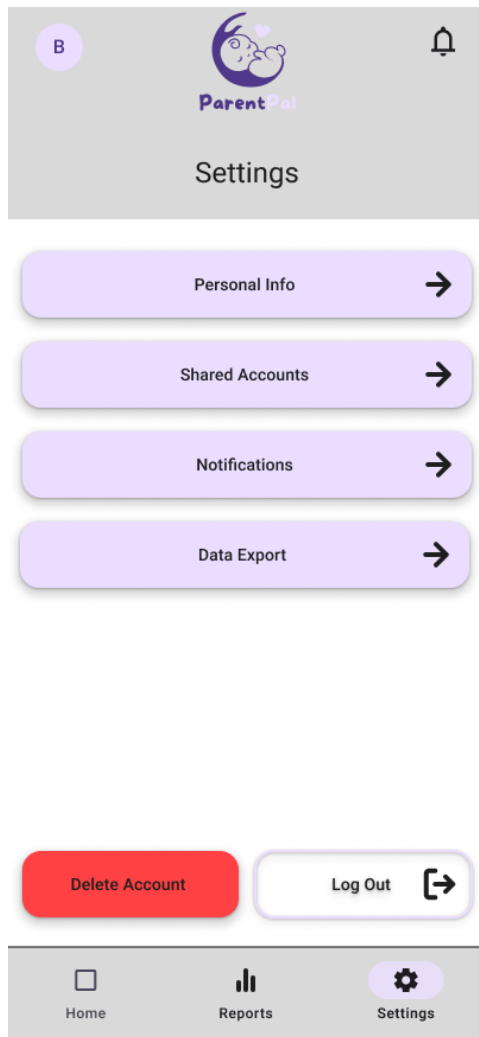
Avg feeding times



4.6 Settings - User to create, update and delete profile information/functions

Parent

Babysitter



4.6.1.1 Settings - Parent Personal Information

User can add new child, update password

Parent Pal

Personal Info

Edit

Personal Data

Email
braeden.lyman1@mohawkcollege.ca

Current Password

Your Little Ones

J Jack (< 1 Month)
2025/02/23

+ Add

Change Password

New Password
Min 8 characters

Confirm Password
Repeat password

Save Changes

Home Reports Settings

Adding in new child

←

Add your baby

B Baby (0-12 months)

T Toddler (1-3 years)

P Preschool (3-5 years)

G Gradeschool (5-12 years)

Next

Filling out child info

←

Tell us more about your toddler

Name
Alex

Date of Birth
2023/02/23

Gender
Female

Add

Personal info after child added

Parent Pal

Personal Info

Personal Data

Email
braeden.lyman1@mohawkcollege.ca

Current Password

Your Little Ones

J Jack (< 1 Month)
2025/02/23

A Alex (2 Years)
2023/02/23

+ Add

Change Password

New Password
Min 8 characters

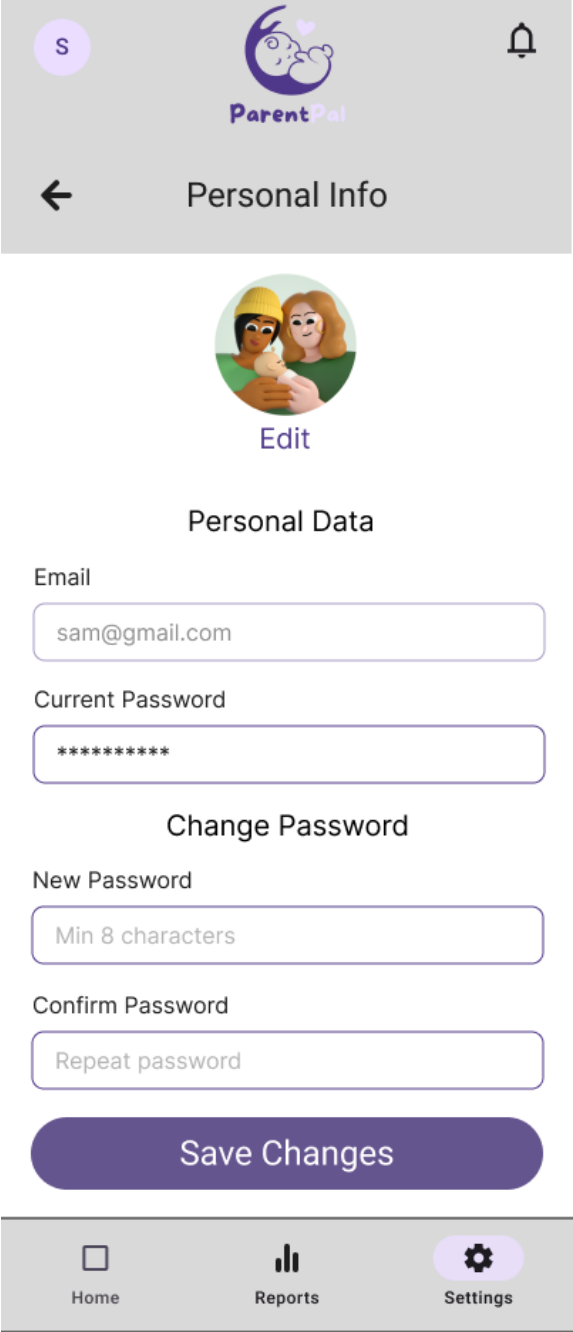
Confirm Password
Repeat password

Save Changes

Home Reports Settings

4.6.1.2 Settings - Babysitter Personal Information

User can update password



ParentPal

Personal Info

Edit

Personal Data

Email

sam@gmail.com

Current Password

Change Password

New Password

Min 8 characters

Confirm Password

Repeat password

Save Changes

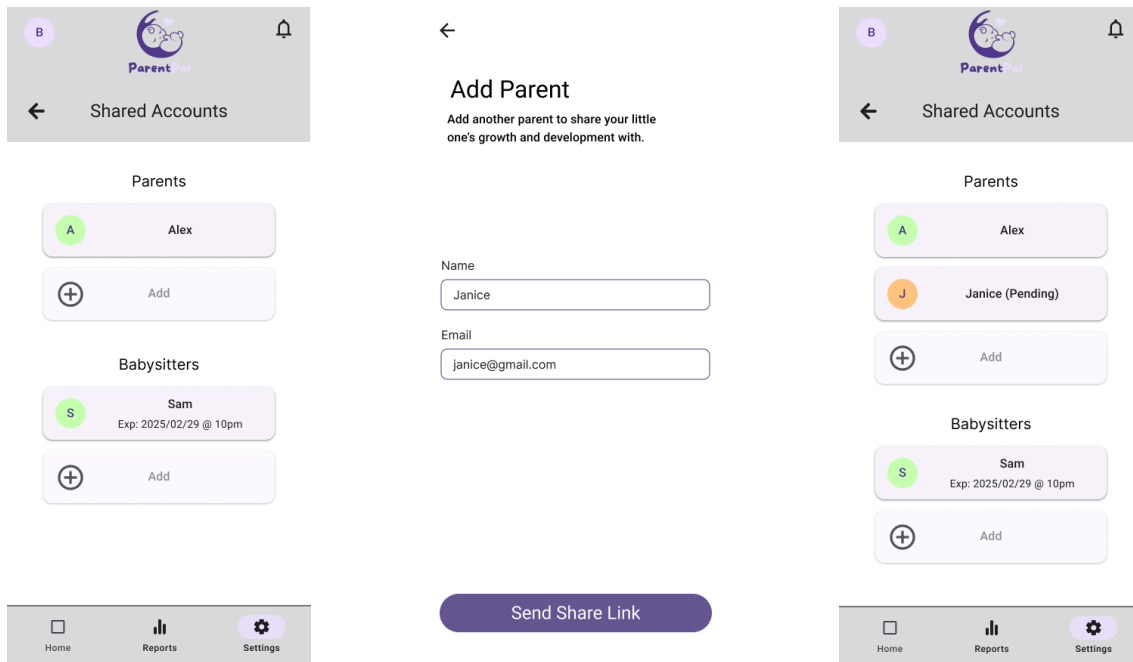
Home Reports Settings

4.6.2.1 Settings - Parent Shared Accounts

Shared accounts
dashboard

Adding new parent

Dashboard after parent
added



Add babysitter w/ expiry or no expiry

←

Add Babysitter

Add a babysitter to share your little one's schedule with and allow them to add notes, photos, and assign them tasks.

Name
Breanne

Email
breanne@gmail.com

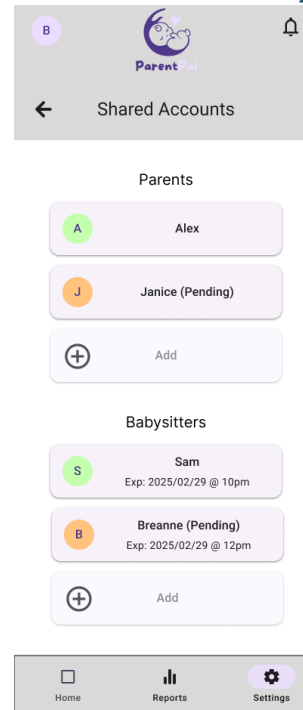
Add Expiry ☒

Start Date & Time
2025/02/29 @ 9am

End Date & Time
2025/02/29 @ 12pm

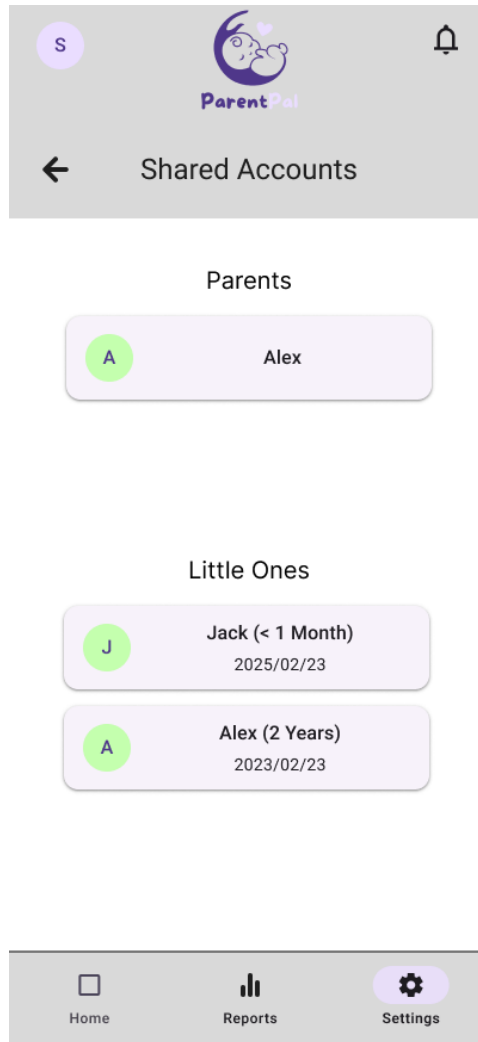
Send Share Link

Dashboard after new babysitter



4.6.2.2 Settings - Babysitter Shared Accounts

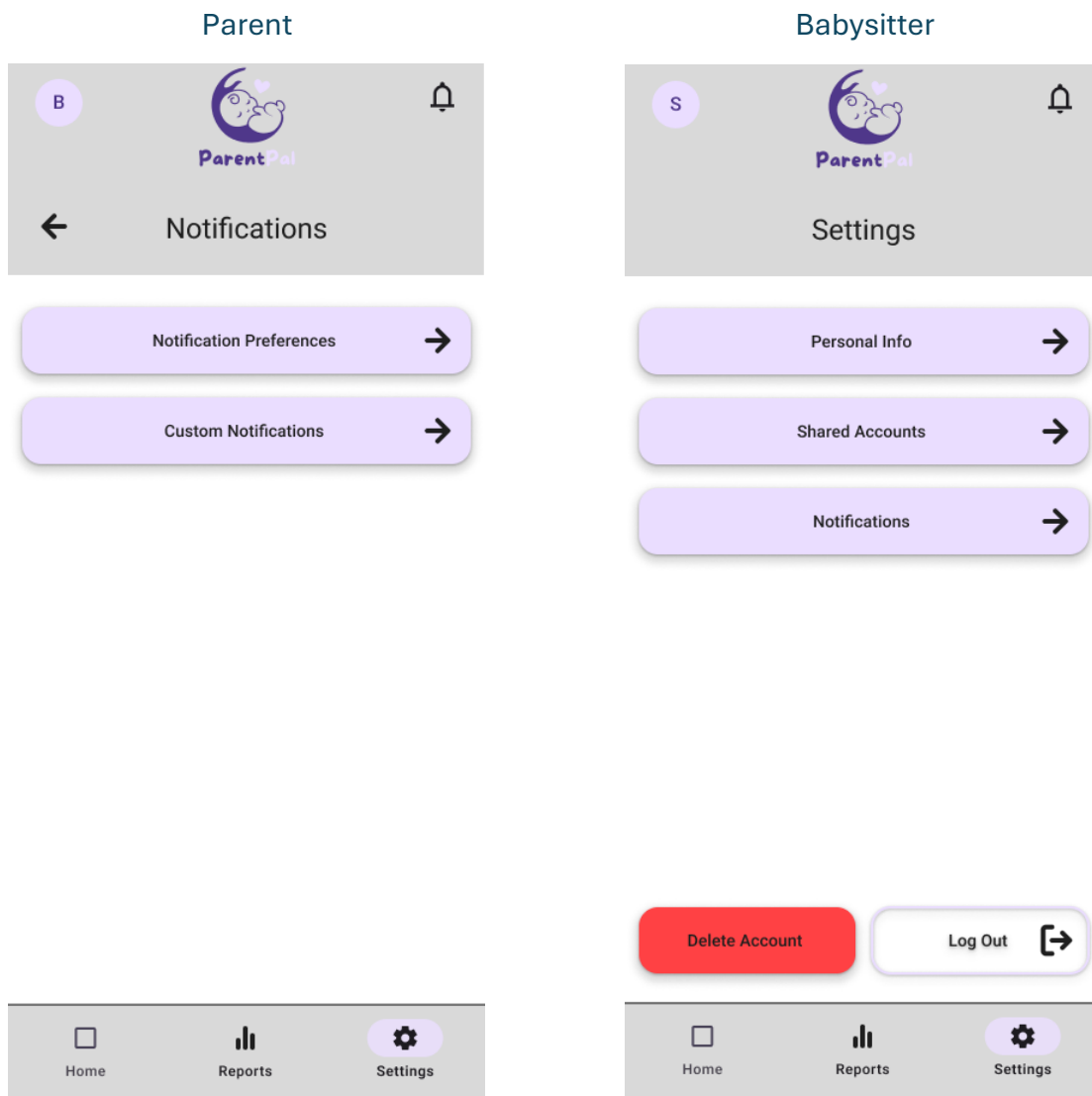
- Shows the accounts the babysitter has limited access to. Shows the parent and parent's associated children



4.6.3 Settings - Notifications/Custom Notifications

- Parent account has notification preferences as well as being able to add in custom notifications

- Babysitter account only has notification preferences, which is the same view as the settings page

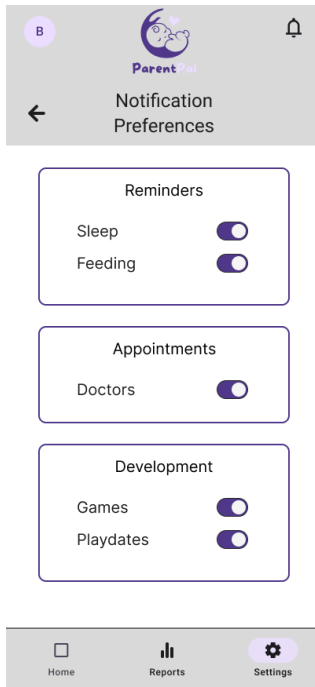


4.6.3.1 Settings - Parent Notifications/Custom Notifications

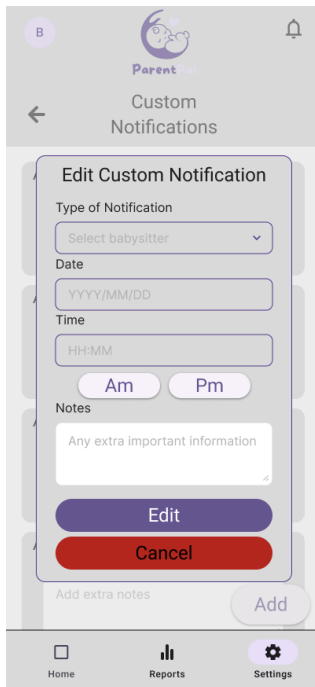
Notification preferences

Custom notifications view

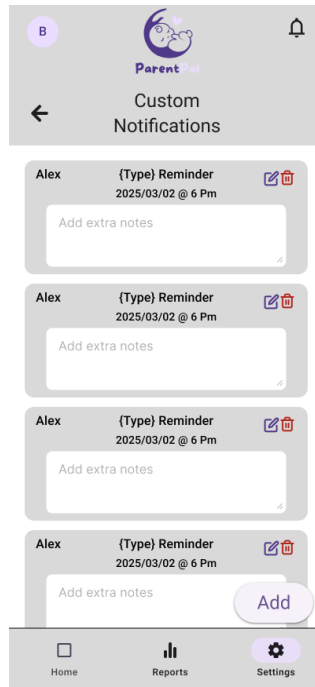
Adding custom notification



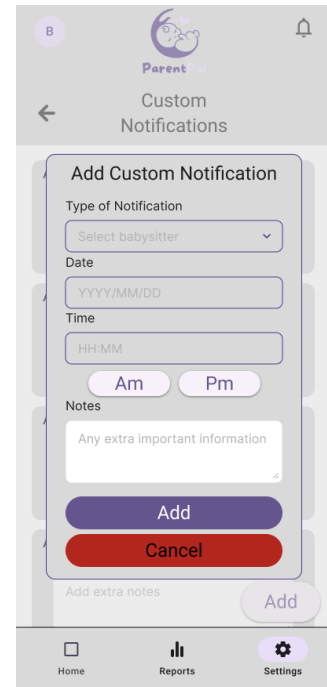
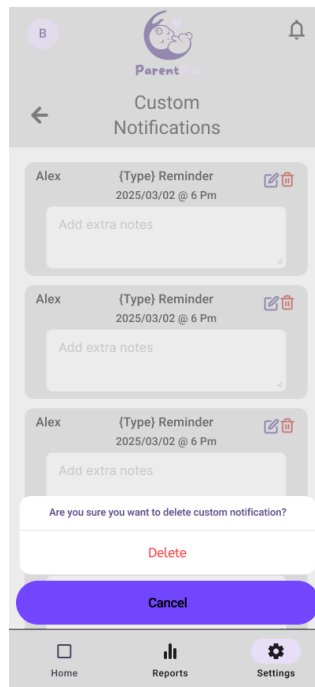
Editing custom notification



4.6.3.2 Settings - Babysitter Notifications

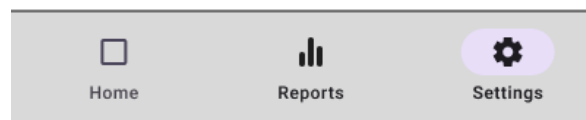
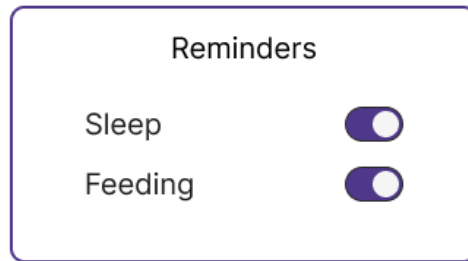
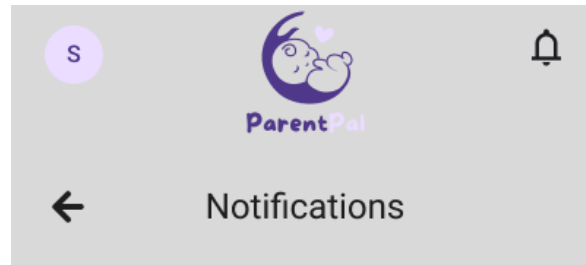


Deleting custom notification



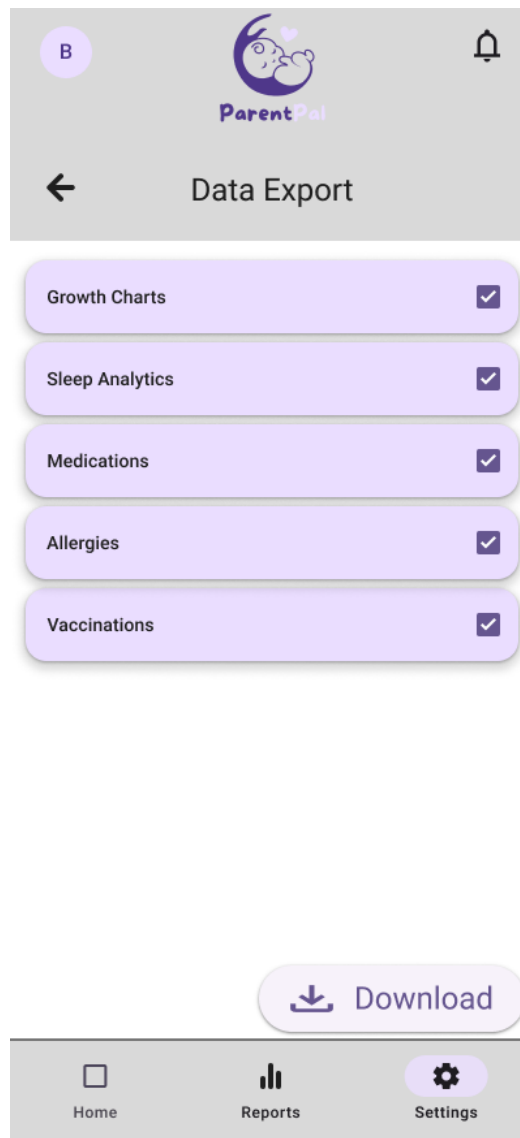
Notification view on phone





4.6.4 Settings - Data Export

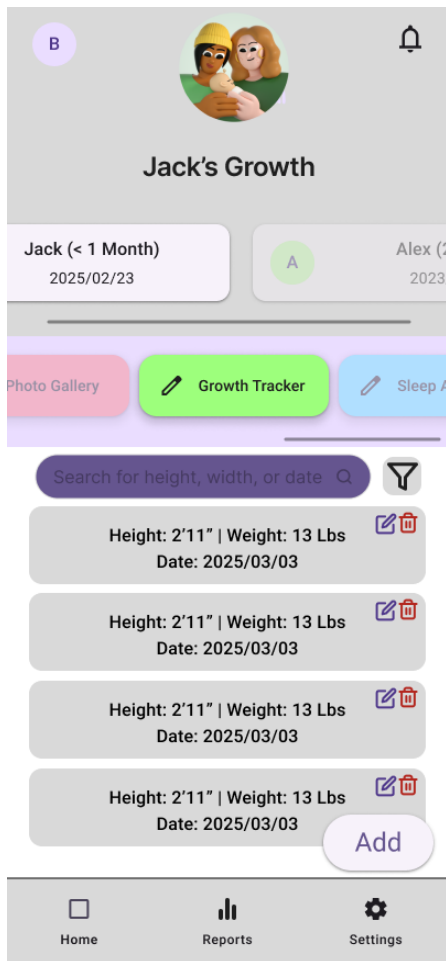
- User can choose which sections will be exported to a PDF file by checking/unchecking each checkbox and clicking the download button
- PDF will be formatted professionally



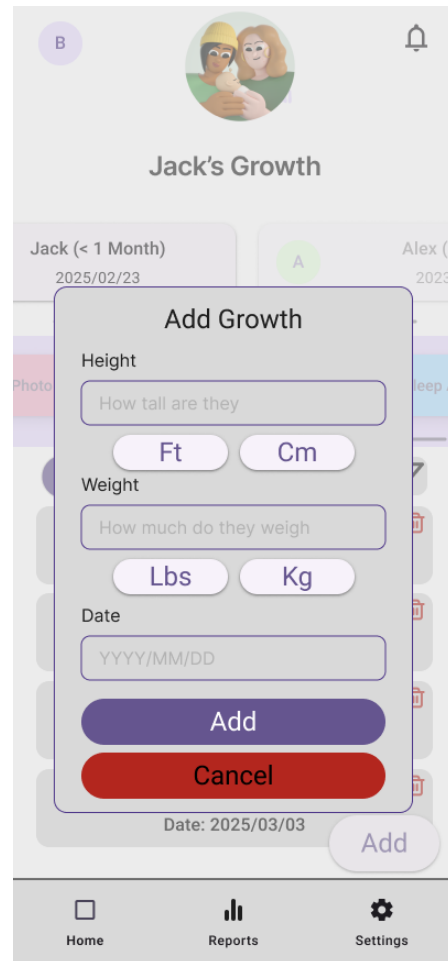
4.7 Growth Tracker

- Tracks the growth of the child
- Only parent can view, add, filter, edit, and delete growth entries (only showing view and add as edit and delete are very similar)

Growth tracker dashboard

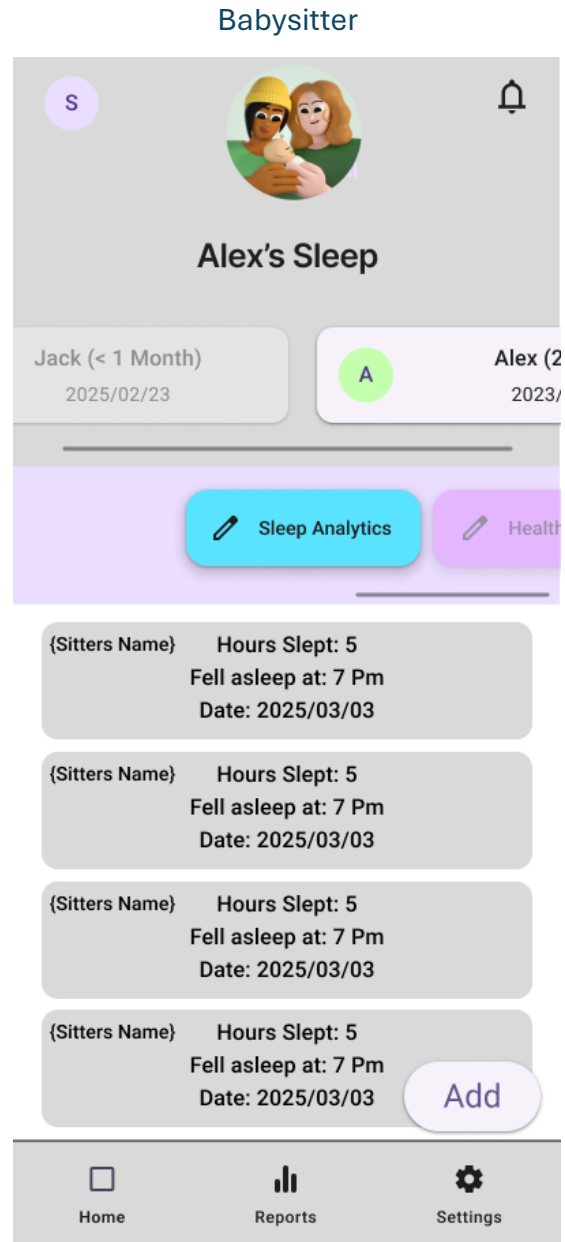
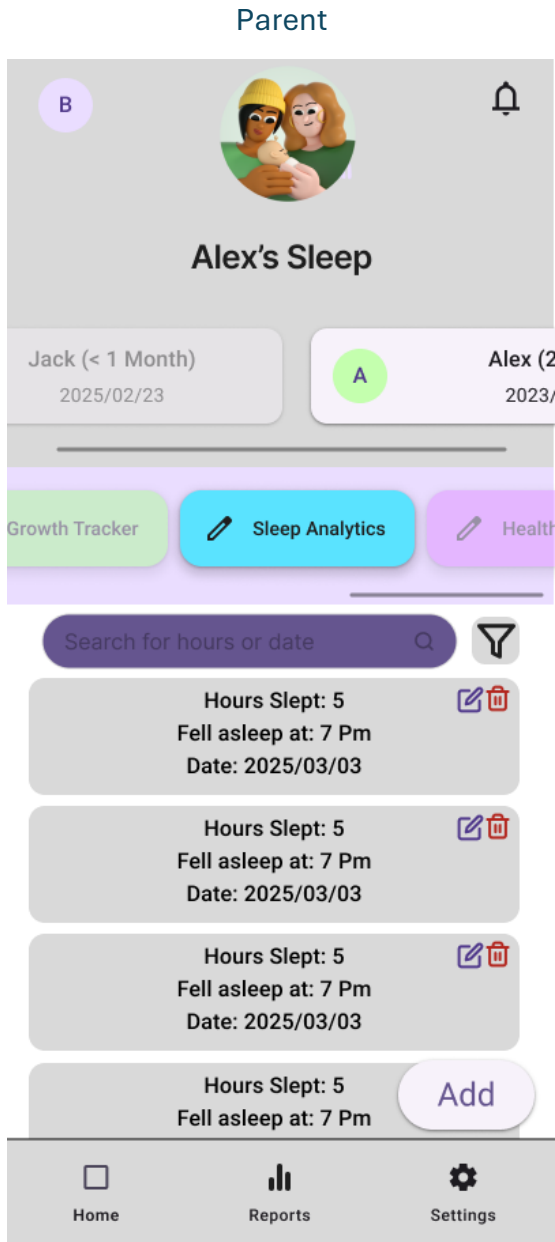


Adding new growth record



4.8 Sleep Analytics

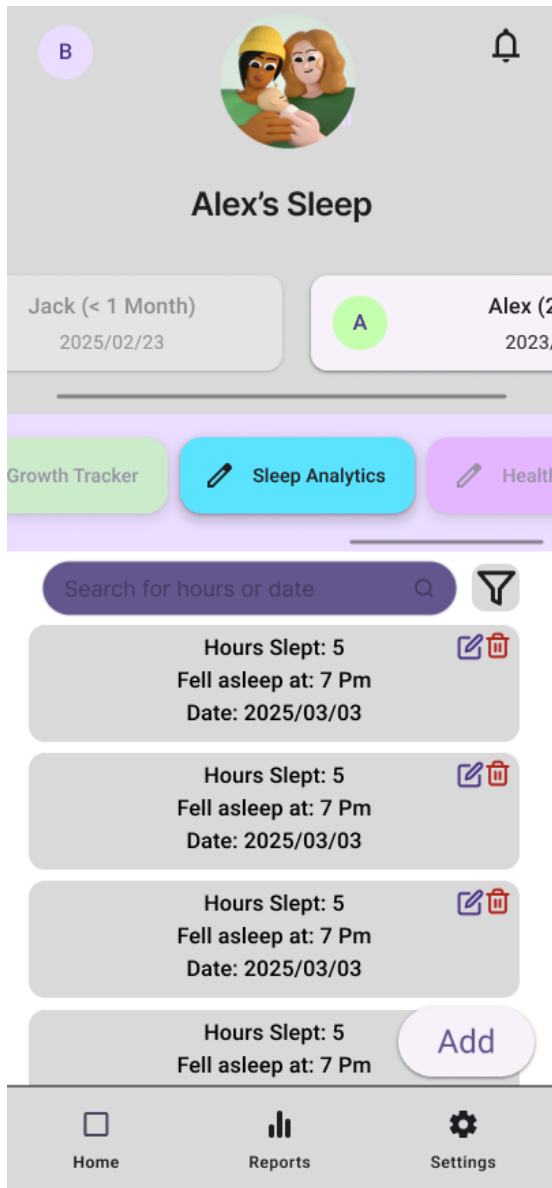
- Tracks sleep analytics of the child



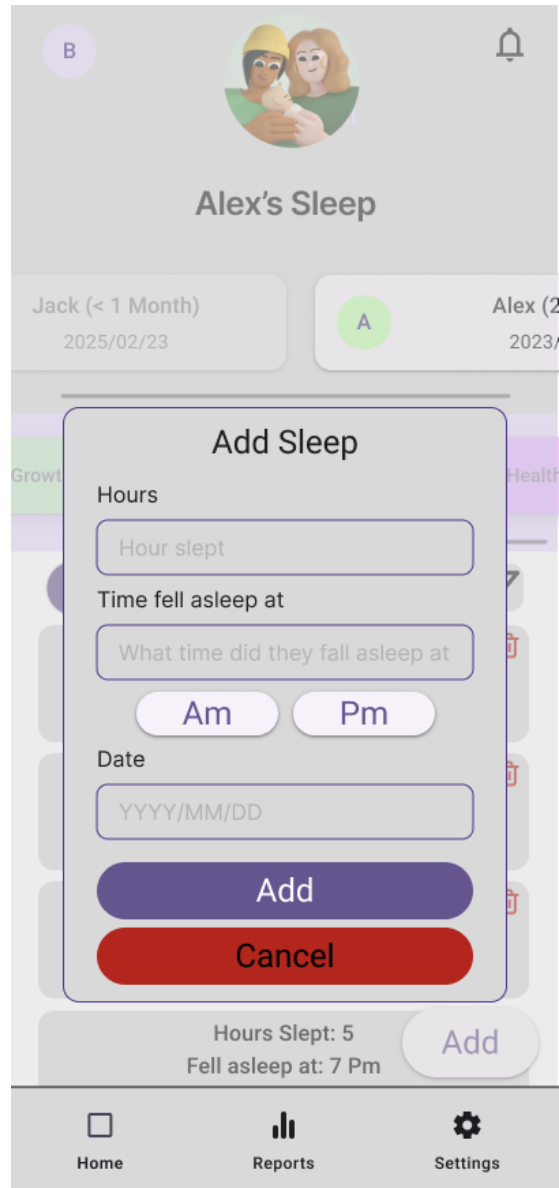
4.8.1 Sleep Analytics - Parent

- Parent has view, add, filter, edit, delete capabilities (only showing view and add as edit and delete are very similar)

Sleep analytics dashboard



Add new sleep record

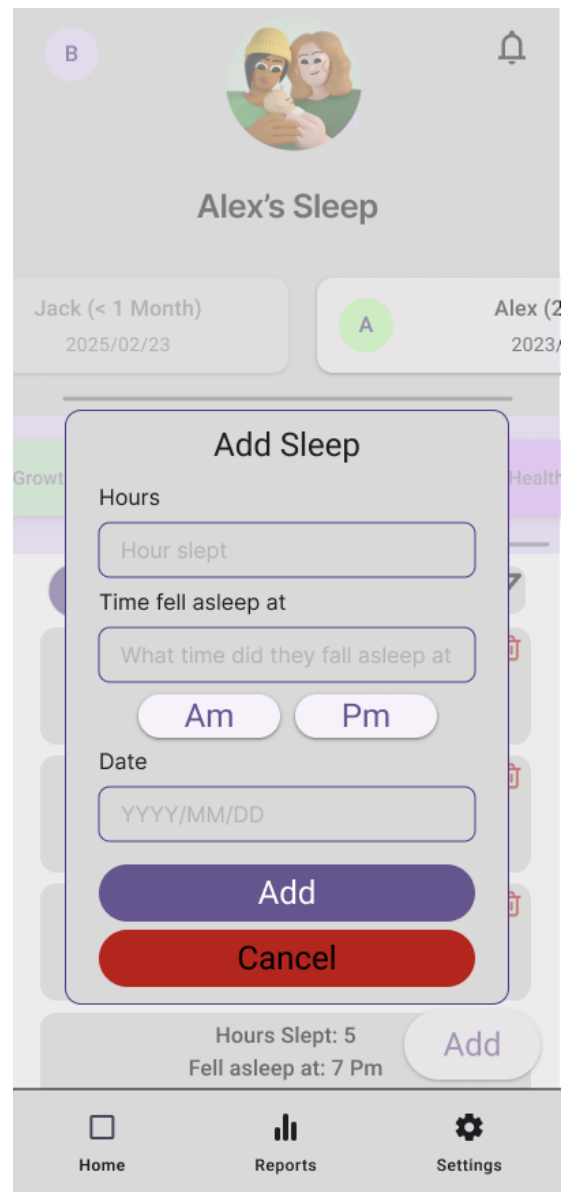
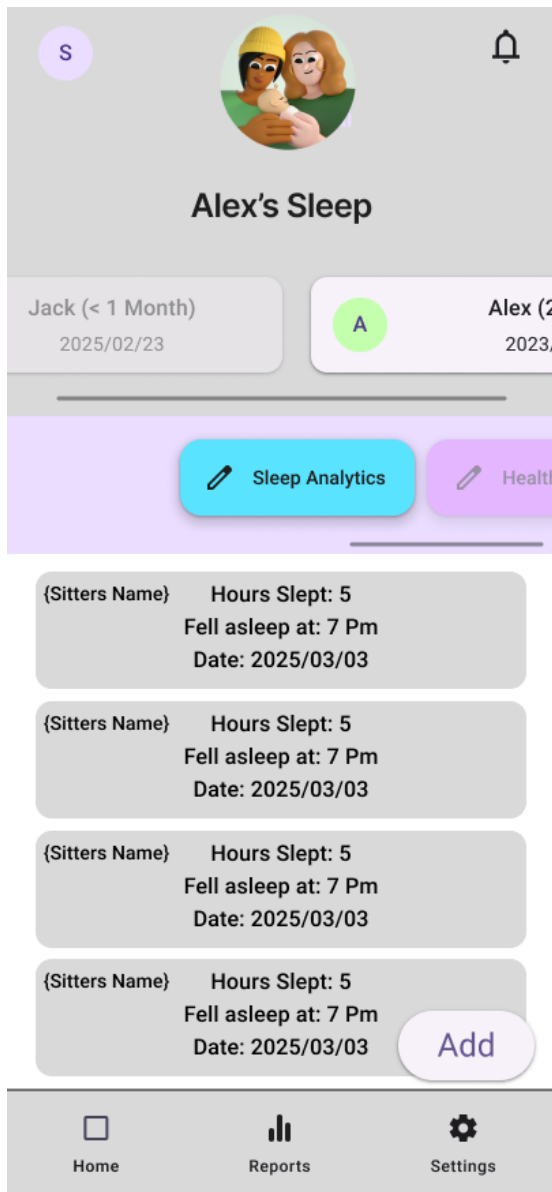


4.8.2 Sleep Analytics - Babysitter

- Babysitter has view and add capabilities

Sleep analytics dashboard

Add new sleep record

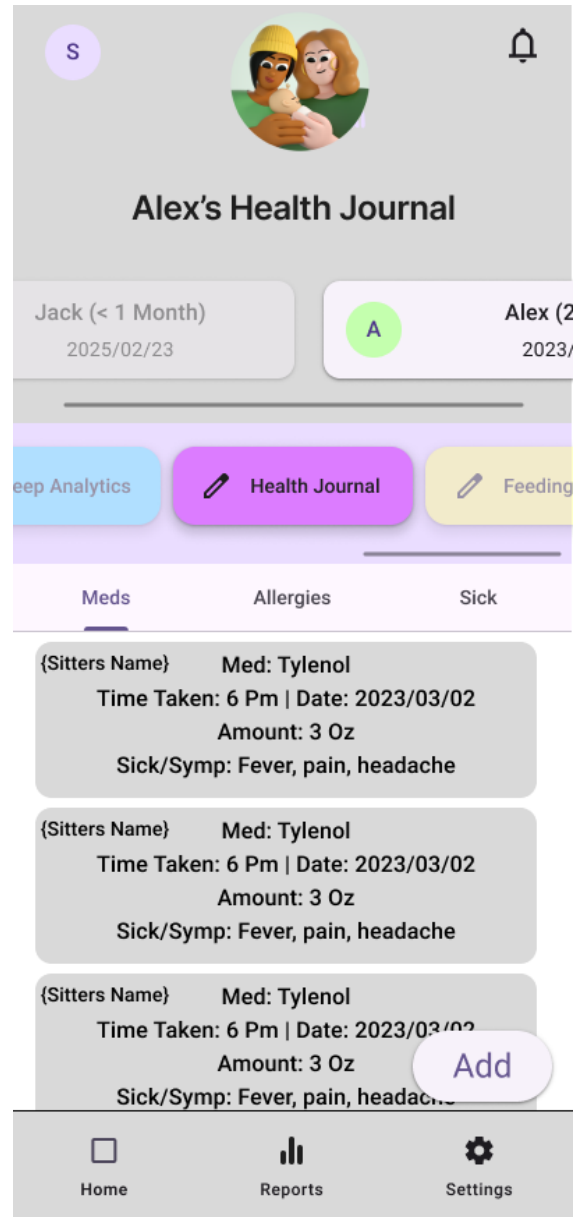
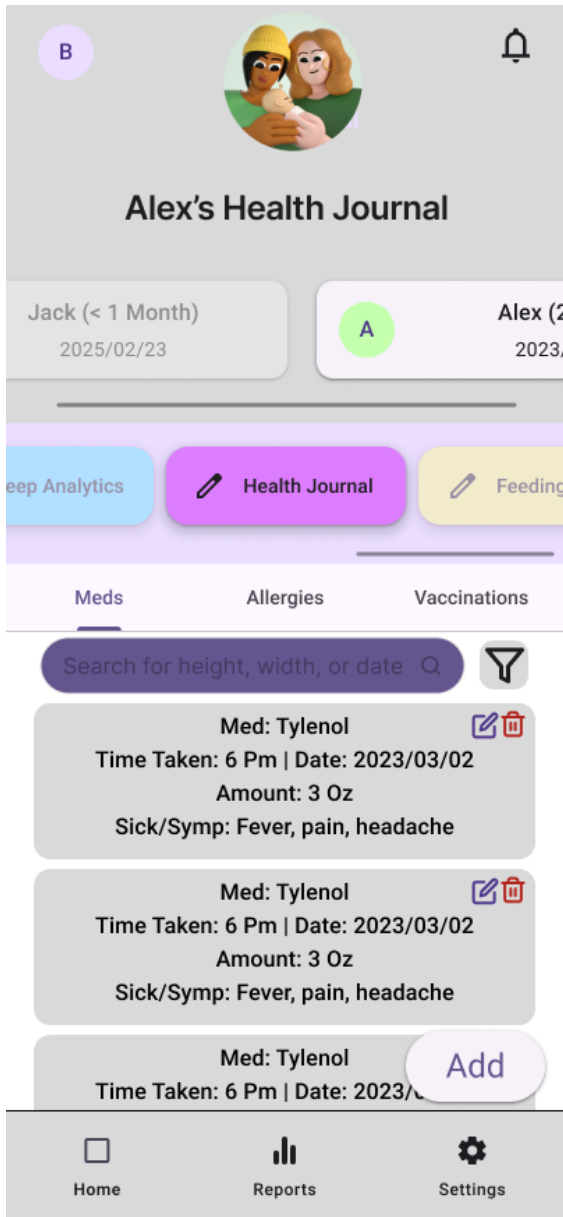


4.9 Health Journal

- Tracks health metrics of the child (has a nav bar with other sections i.e. Allergies, Vaccinations, Sick Days)

Parent

Babysitter



4.9.1 Health Journal - Parent

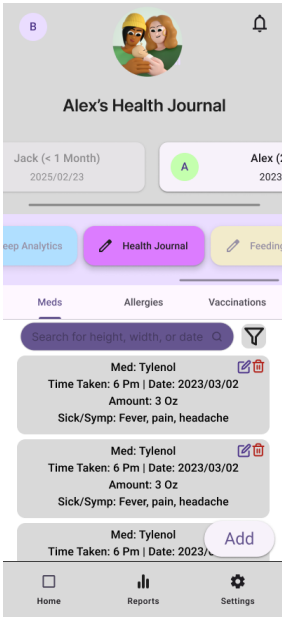
- Parent has view, add, filter, edit, delete capabilities (only showing view and add as edit and delete are very similar)
- Parent can log Med, Allergies, Vaccinations and Sick Days

Med view

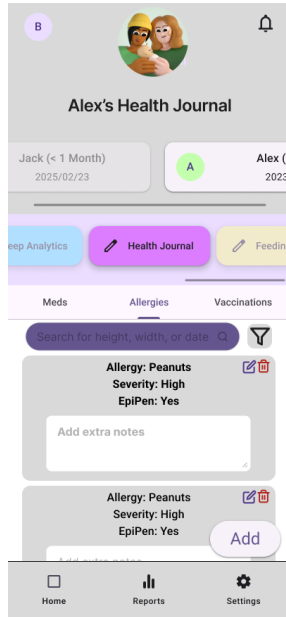
Allergies view

Vaccinations view

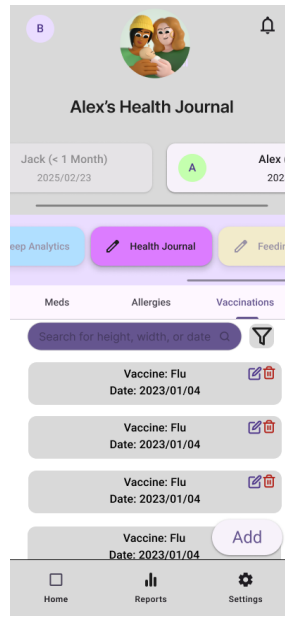
Sick Days view



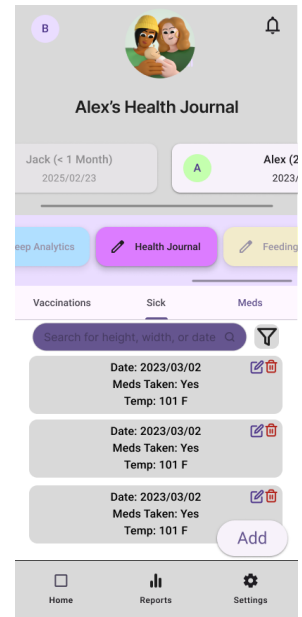
Add meds



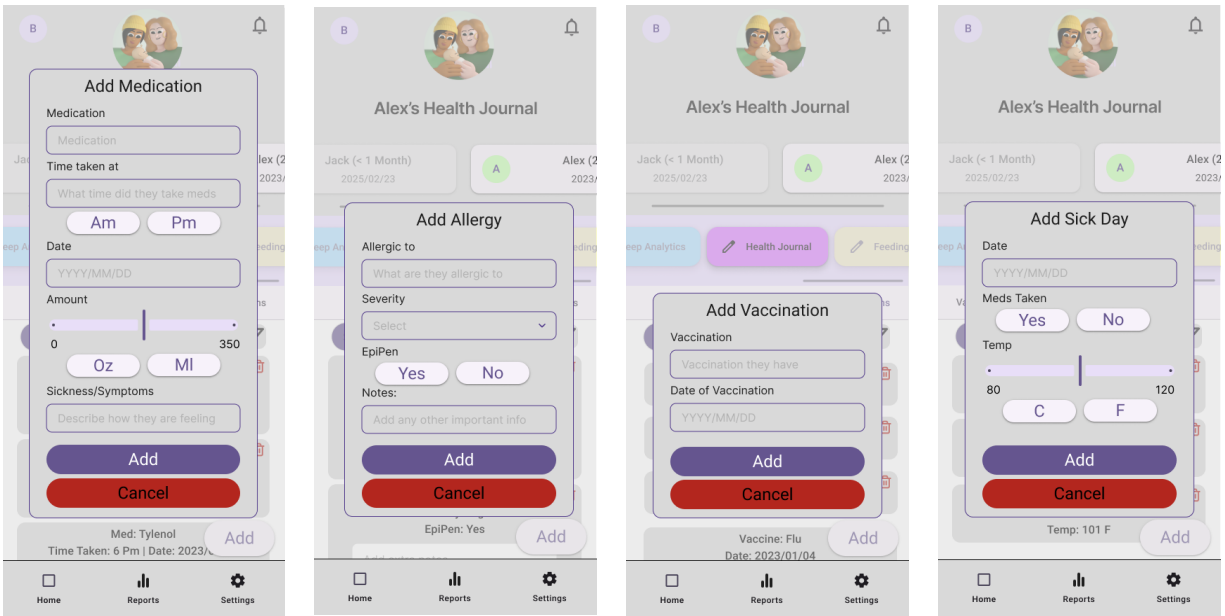
Add allergies



Add vaccinations

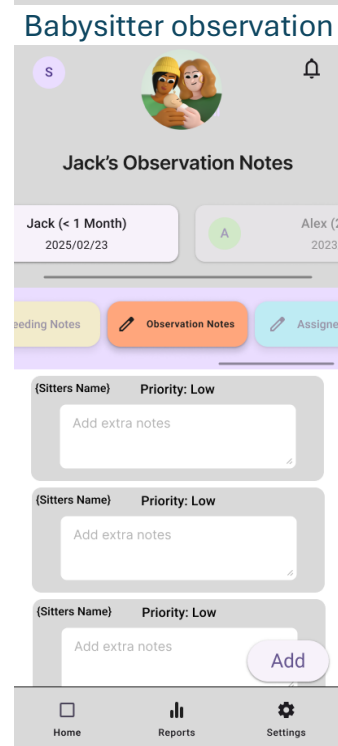
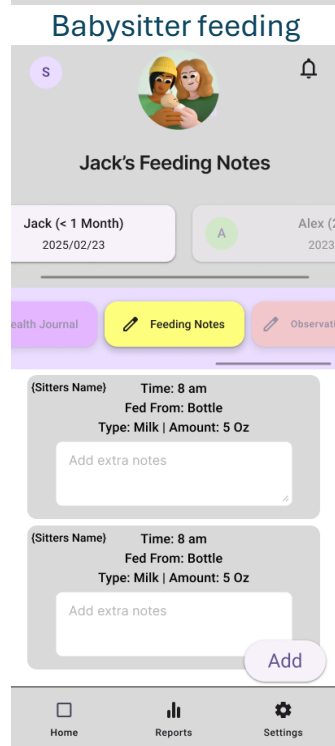
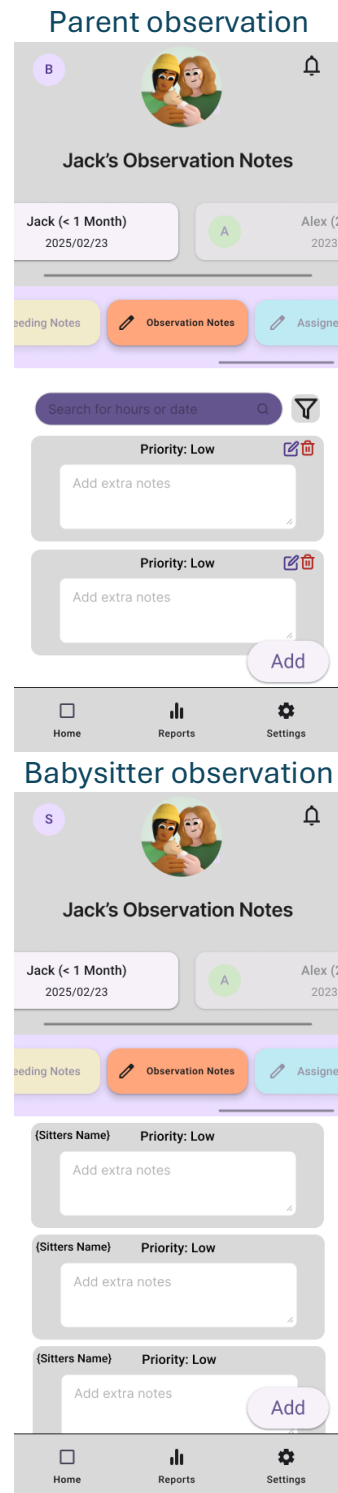
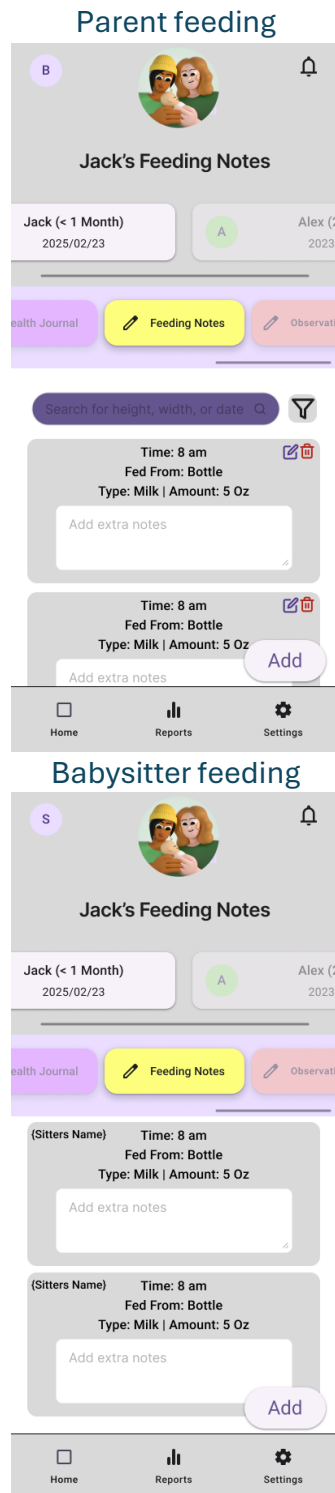


Add sick days



4.10 Notes Observation/Feeding

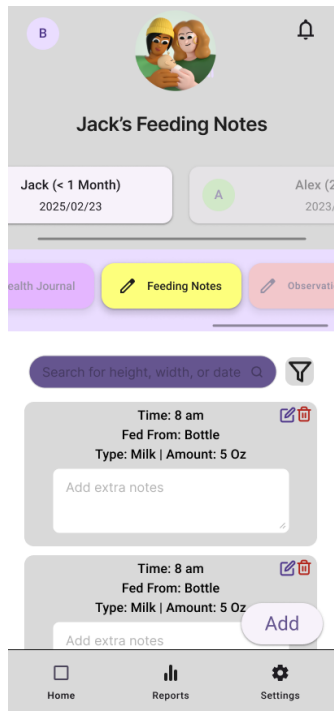
- Tracks observation and feeding notes of the child



4.10.1 Notes Observation/Feeding - Parent

- Parent has view, add, filter, edit, delete capabilities (only showing view and add as edit and delete are very similar)

Feeding dashboard



The Feeding dashboard for Jack shows a header with a profile icon, a bell, and the title "Jack's Feeding Notes". Below the header, there are two tabs: "Jack (< 1 Month)" and "Alex (2023/02/23)". The main content area has three buttons: "Health Journal", "Feeding Notes" (highlighted in yellow), and "Observation Notes". Below the buttons is a search bar with the placeholder "Search for height, width, or date". The main content area displays two feed records. Each record shows the time (8 am), the source (Fed From: Bottle), the type (Milk), and the amount (5 Oz). There is a text input field for "Add extra notes" and an "Add" button. The bottom navigation bar has three icons: Home, Reports, and Settings.

Jack's Feeding Notes

Jack (< 1 Month) 2025/02/23

Health Journal Feeding Notes Observation Notes

Search for height, width, or date

Time: 8 am
Fed From: Bottle
Type: Milk | Amount: 5 Oz
Add extra notes

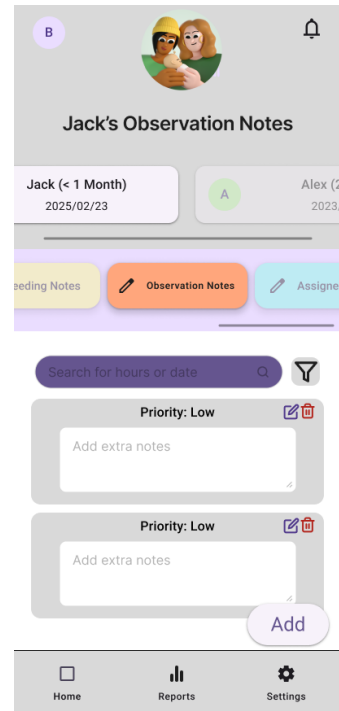
Time: 8 am
Fed From: Bottle
Type: Milk | Amount: 5 Oz
Add extra notes

Add

Home Reports Settings

Add new feed record

Observation dashboard



The Observation dashboard for Jack shows a header with a profile icon, a bell, and the title "Jack's Observation Notes". Below the header, there are two tabs: "Jack (< 1 Month)" and "Alex (2023/02/23)". The main content area has three buttons: "Feeding Notes", "Observation Notes" (highlighted in orange), and "Assignment Notes". Below the buttons is a search bar with the placeholder "Search for hours or date". The main content area displays two observation records. Each record shows the priority (Low) and a text input field for "Add extra notes". There is an "Add" button. The bottom navigation bar has three icons: Home, Reports, and Settings.

Jack's Observation Notes

Jack (< 1 Month) 2025/02/23

Feeding Notes Observation Notes Assignment Notes

Search for hours or date

Priority: Low
Add extra notes

Priority: Low
Add extra notes

Add

Home Reports Settings

Add new observation record

B

Add Feeding

Time fed at

HH:MM AM/PM

Fed from

Select

Type

Select

Oz

MI

Amount

0350

Notes:

Add any other important info

Add

Cancel

Type: Milk | Amount: 5 Oz

Add extra notes

Add

Home

Reports

Settings

S

Jack's Observation Notes

Jack (< 1 Month)

2025/02/23

A

Alex (2023/02/23)

Add Observation

Priority Level

Select

Notes

Describe what you notice

Add

Cancel

Add extra notes

Add

Home

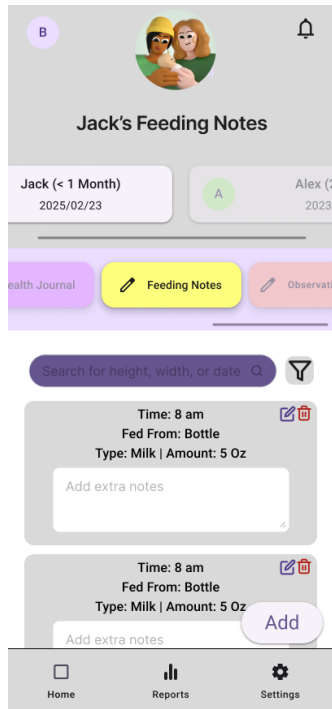
Reports

Settings

4.10.2 Notes Observation/Feeding - Babysitter

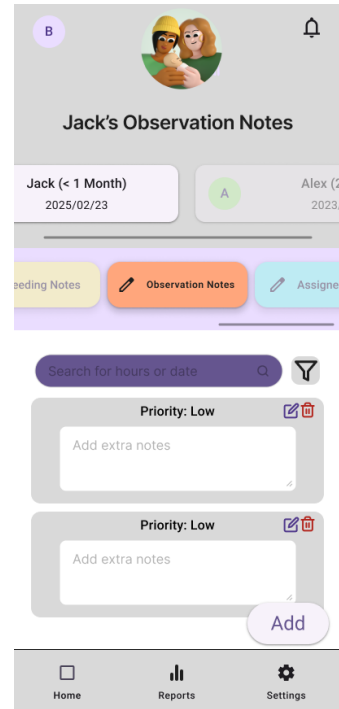
- Babysitter has view and add capabilities for feeding and observation notes

Feeding dashboard



The Feeding dashboard for Jack's Feeding Notes shows a header with a profile icon, a bell, and the title. Below the header, there are two tabs: 'Jack (< 1 Month)' and 'Alex (2023/02/23)'. A navigation bar at the bottom of the header contains 'Health Journal', 'Feeding Notes' (highlighted), and 'Observations'. The main content area has a search bar 'Search for height, width, or date' and a list of feeding notes. Each note displays 'Time: 8 am', 'Fed From: Bottle', and 'Type: Milk | Amount: 5 Oz', followed by an 'Add extra notes' field. An 'Add' button is visible at the bottom right of the list. The bottom navigation bar includes 'Home', 'Reports', and 'Settings'.

Observation dashboard



The Observation dashboard for Jack's Observation Notes shows a header with a profile icon, a bell, and the title. Below the header, there are two tabs: 'Jack (< 1 Month)' and 'Alex (2023/02/23)'. A navigation bar at the bottom of the header contains 'Feeding Notes', 'Observation Notes' (highlighted), and 'Assignments'. The main content area has a search bar 'Search for hours or date' and a list of observation notes. Each note displays 'Priority: Low' and 'Add extra notes' field. An 'Add' button is visible at the bottom right of the list. The bottom navigation bar includes 'Home', 'Reports', and 'Settings'.

Add new feed record

The 'Add Feeding' modal form is displayed over a background showing a user profile for 'B' and a list of feeding records. The form includes the following fields and controls:

- Time fed at:** A text input field with a placeholder 'HH:MM AM/PM'.
- Fed from:** A dropdown menu with 'Select' as the current option.
- Type:** A dropdown menu with 'Select' as the current option.
- Amount:** A range slider between 0 and 350, with a current value of 5 Oz.
- Notes:** A text input field with a placeholder 'Add any other important info'.
- Buttons:** A purple 'Add' button and a red 'Cancel' button.

Below the modal, a summary bar shows 'Type: Milk | Amount: 5 Oz' and an 'Add' button. The bottom navigation bar contains icons for Home, Reports, and Settings.

Add new observation record

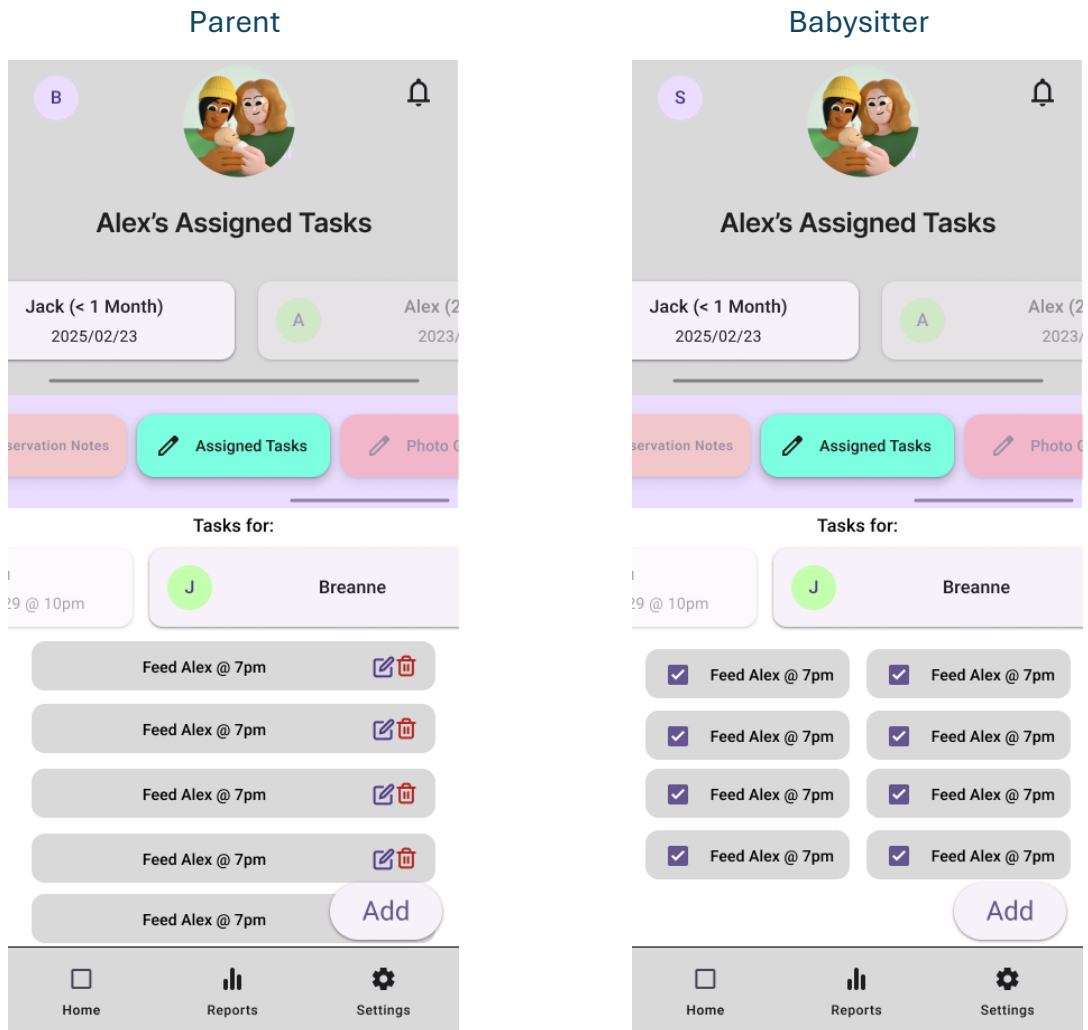
The 'Add Observation' modal form is displayed over a background showing a user profile for 'S' and a list of observation notes. The form includes the following fields and controls:

- Priority Level:** A dropdown menu with 'Select' as the current option.
- Notes:** A text input field with a placeholder 'Describe what you notice'.
- Buttons:** A purple 'Add' button and a red 'Cancel' button.

Below the modal, a summary bar shows 'Add extra notes' and an 'Add' button. The bottom navigation bar contains icons for Home, Reports, and Settings.

4.11 Assigned Tasks

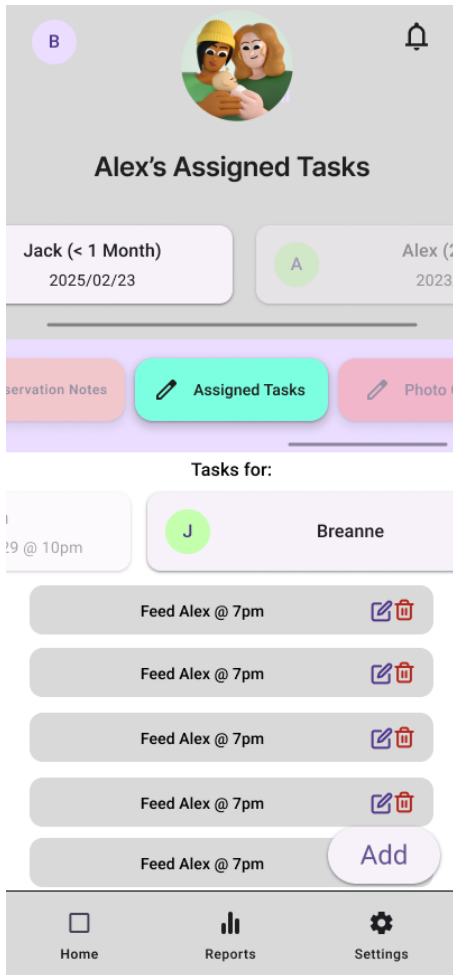
- Tasks that the parent will assign to the babysitter to complete



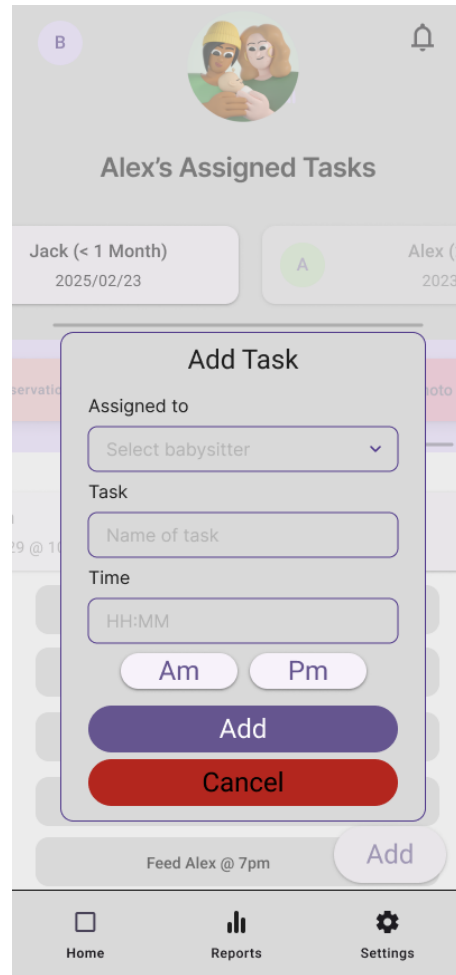
4.11.1 Assigned Tasks - Parent

- Parent assigns tasks to the babysitter
- Parent can view, add, edit and delete tasks (only showing view and add as edit and delete are very similar)

Assigned tasks dashboard



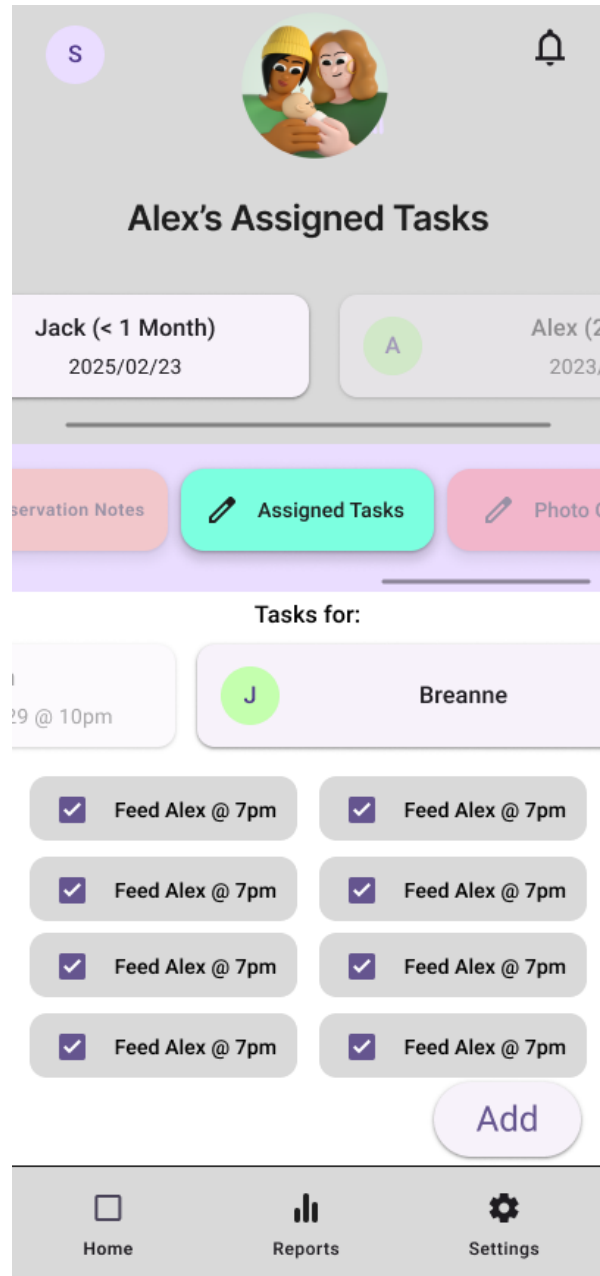
Add new assigned task



4.11.2 Assigned Tasks - Babysitter

- Babysitter can view and complete tasks (By using the checkbox)

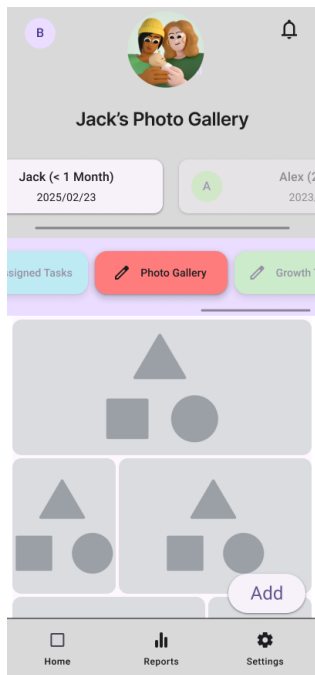
Assigned tasks dashboard with checkboxes (complete/Incomplete)



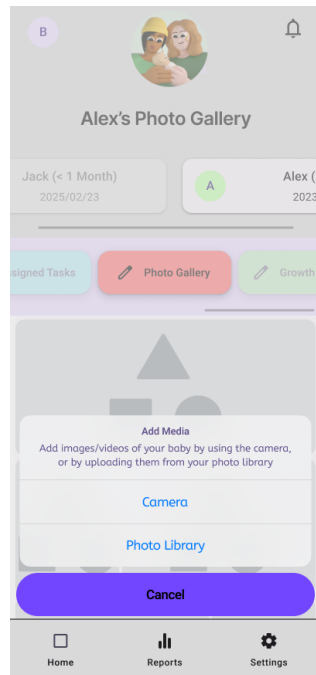
4.12 Media Gallery

- Adding images and videos of the children and displaying them in a gallery
- Parent can view, add, and delete media
- Babysitter can view and add media

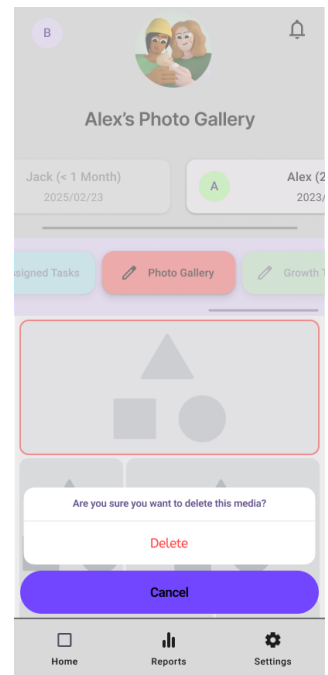
Media view



Add new media



Delete media (Parent)



4.13 Desktop/Landscape View

- Some mockups of how the application will work on Desktop and Landscape view

Sign In Landscape (No Face ID)



Welcome back!

Email

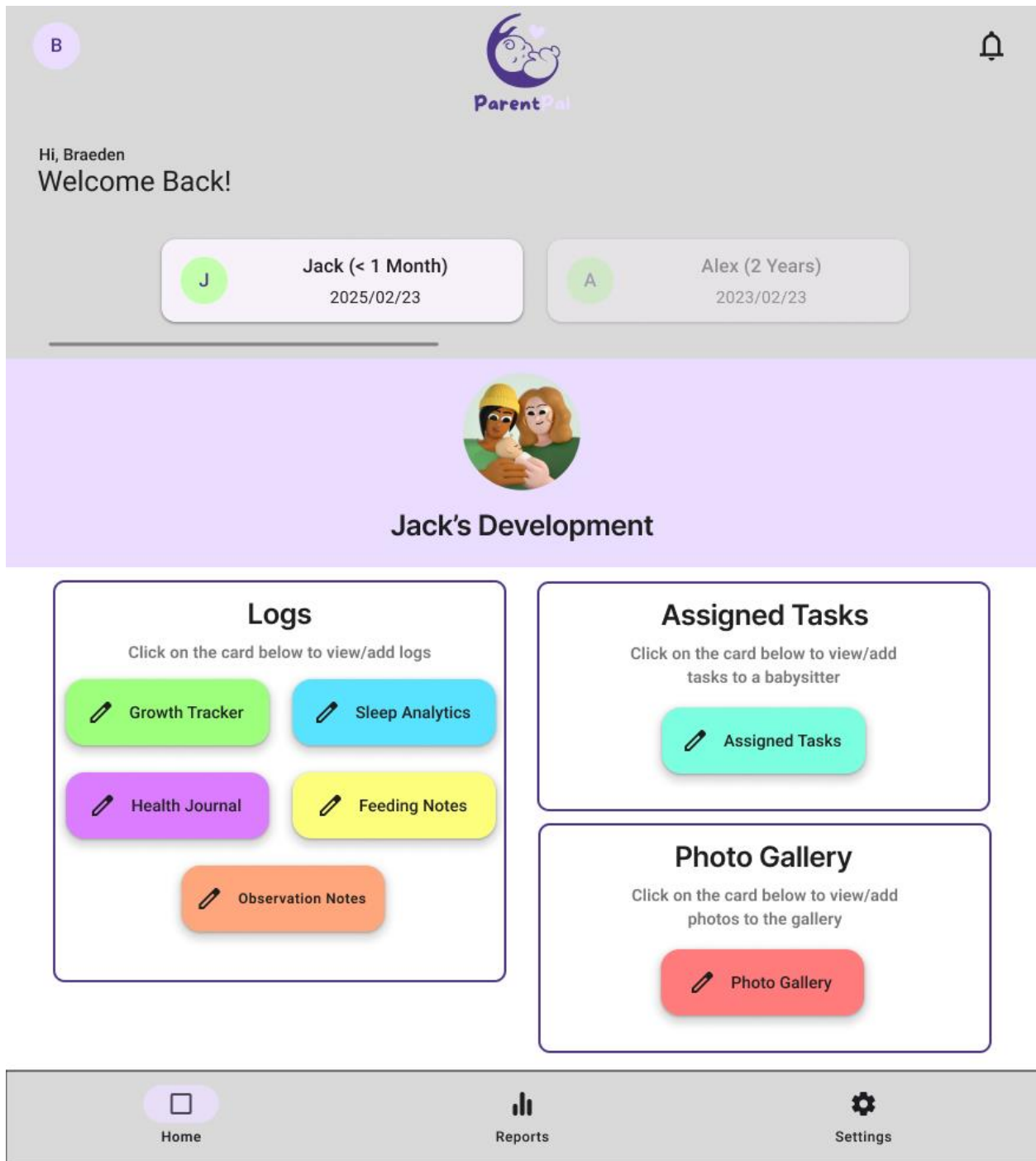
Password

[Forgot Password?](#)

Log In

Are you new here? [Create account](#)

Dashboard Landscape



Sign Up Sequence Landscape



Create an account

Full Name

Email

Password

Create an account

Already have an account? [Log in](#)



Enter the 4-digit code

Your passcode has been sent to your email address
braeden.lyman1@mohawkcollege.ca

0

Resent OTP will be available in 00:12 sec

From Messages
050518

1	2 A B C	3 D E F
4 G H I	5 J K L	6 M N O
7 P Q R S	8 T U V	9 W X Y Z
	0	



Successfully verified

Let's start creating your account so
you can start using the app

Start



What brings you to our app?

P

I am a parent

I am here to keep track and manage my baby's growth and development

B

I am a babysitter

I am here to babysit and stay on track for the baby's schedule and development

Next



Tell us more about yourself



Date of Birth

YYYY/MM/DD

Gender

Select



Next



Tell us more about yourself



Date of Birth

1997/01/04

Gender

Male

Next



Add your baby

B

Baby (0-12 months)

P

Preschool (3-5 years)

T

Toddler (1-3 years)

G

Gradeschool (5-12 years)

Next



Tell us more about your baby

Name

Jack



Date of Birth

2025/02/23

Gender

Male



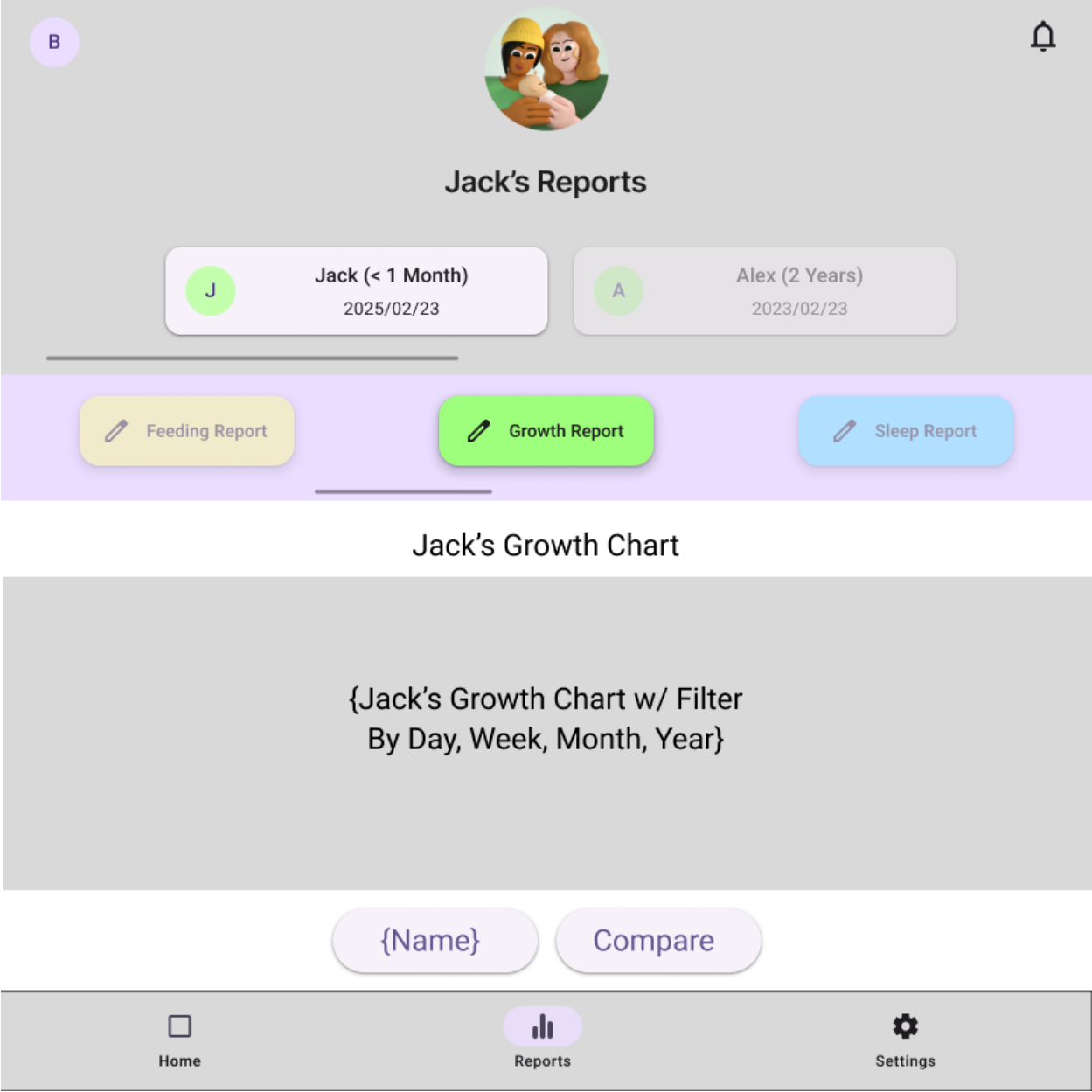
Finish

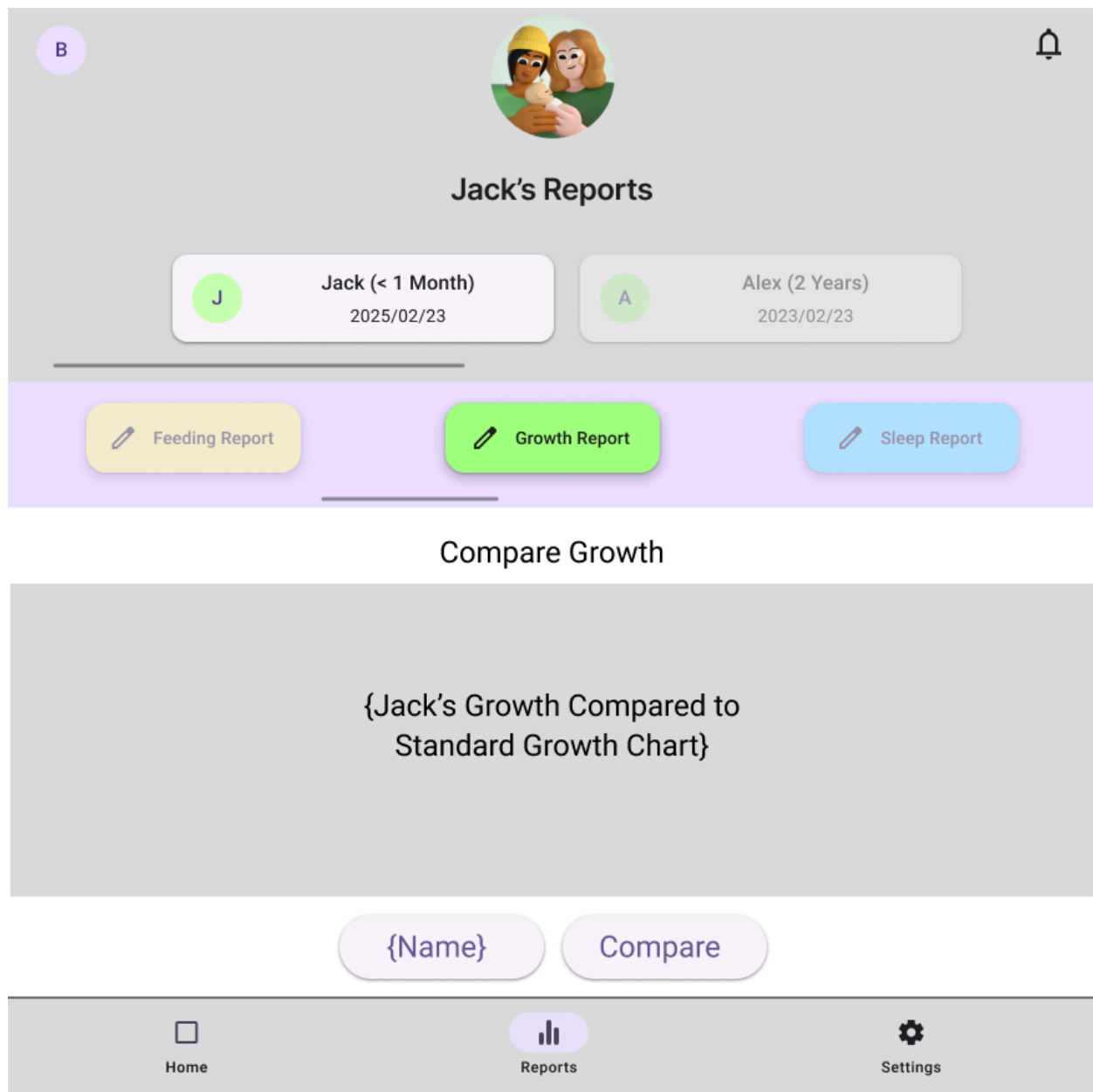


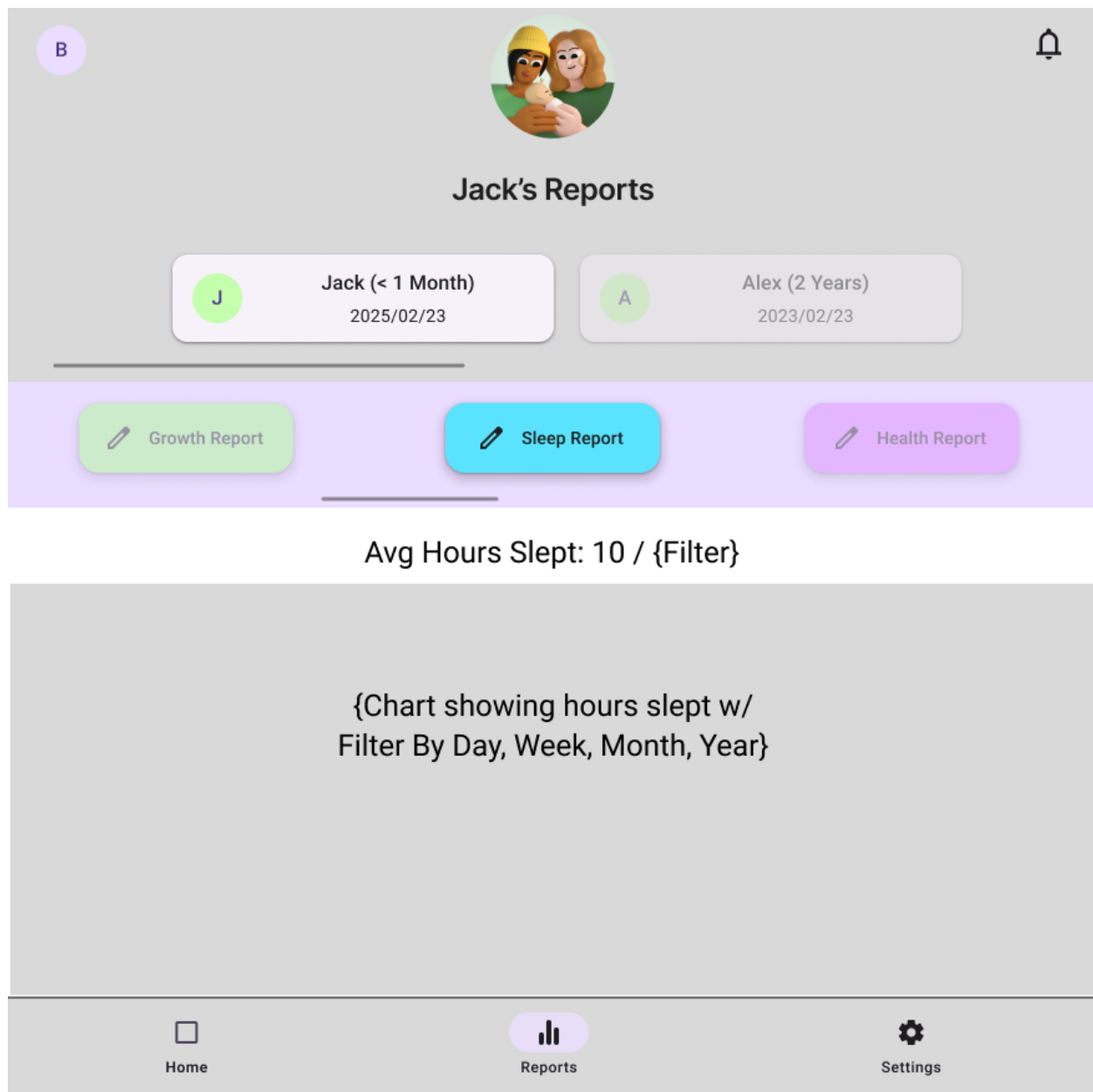
Profile successfully completed

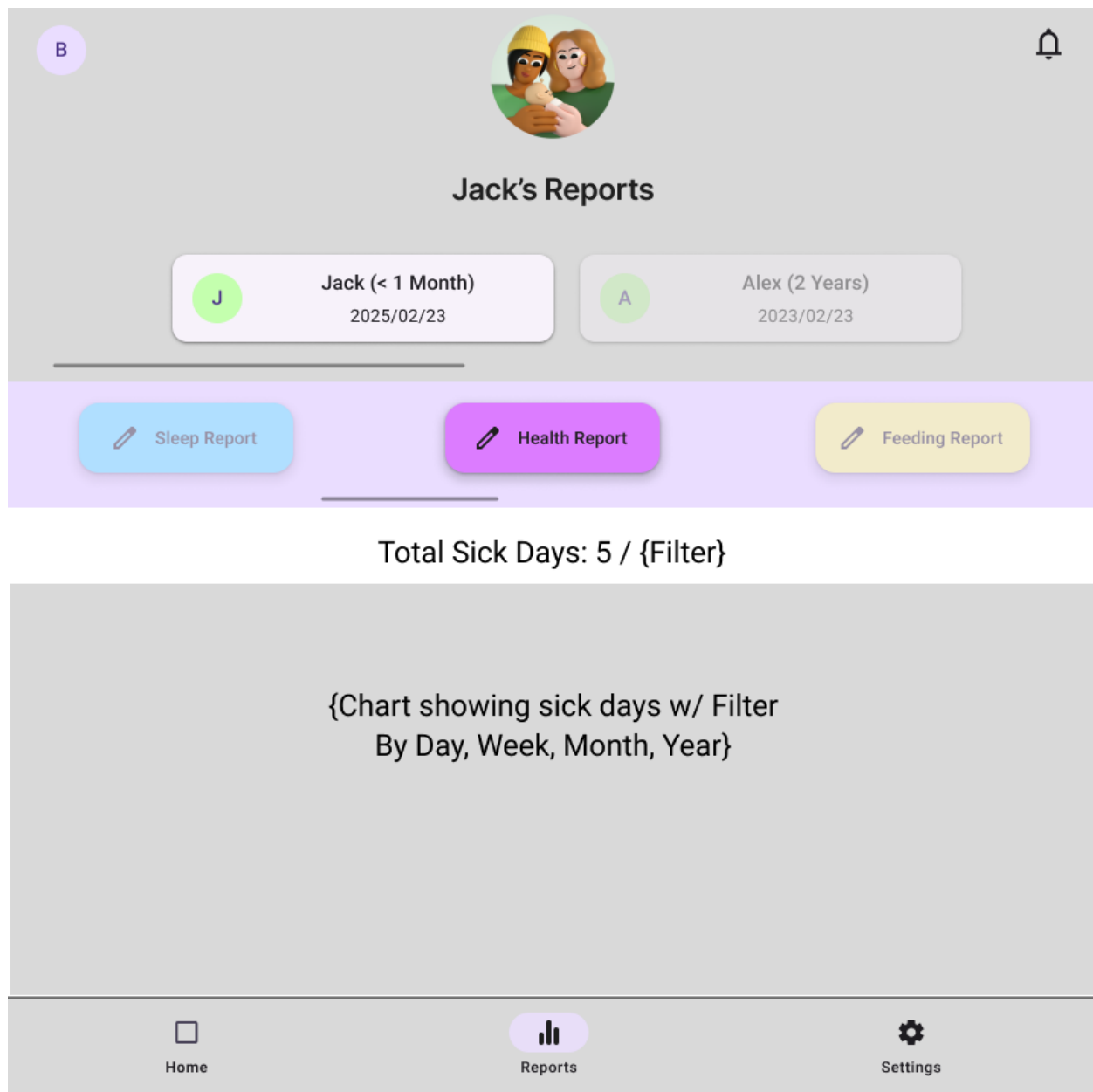
Now you can go to the app and start tracking your baby's development

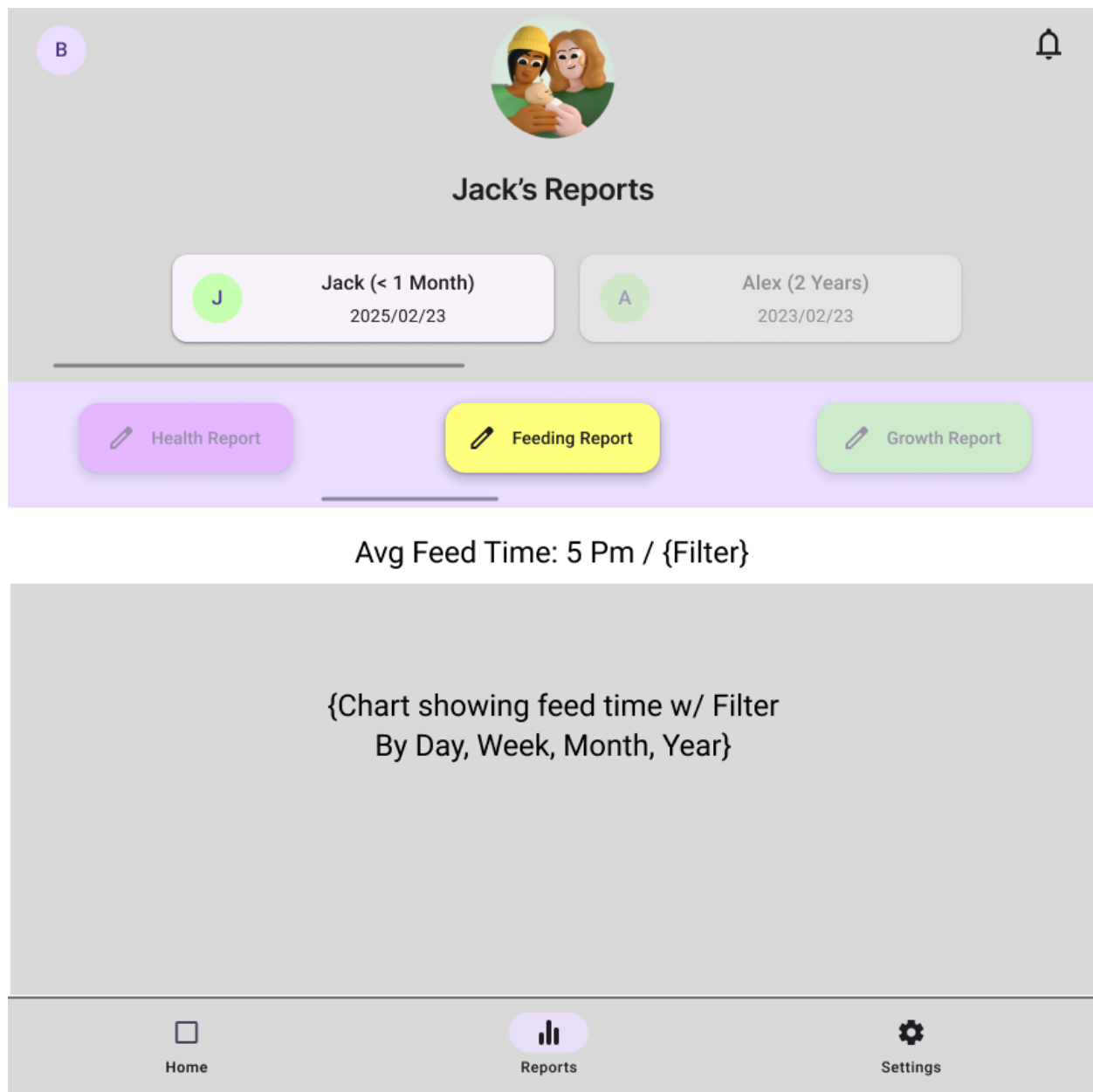
Go to home page



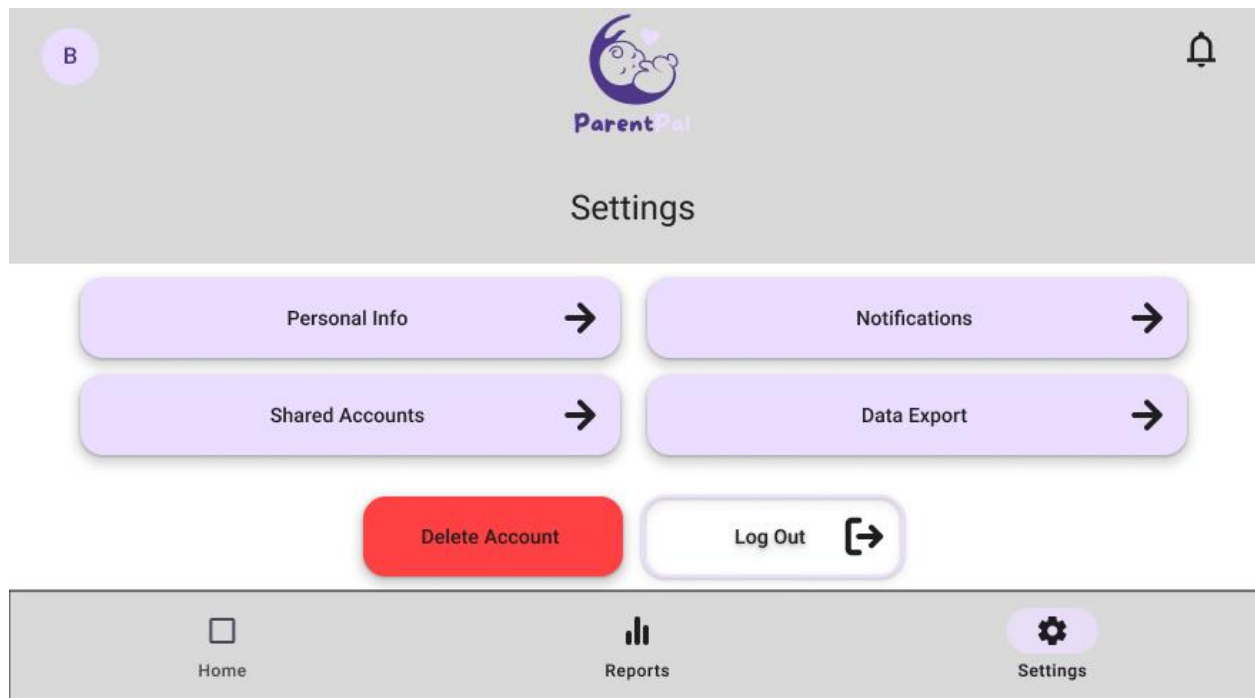








Settings Landscape



B

ParentPal



Personal Info



Edit

Personal Data

Email

braeden.lyman1@mohawkcollege.ca

Current Password

Your Little Ones

J

Jack (< 1 Month)
2025/02/23



Add

Change Password

New Password

Min 8 characters

Confirm Password

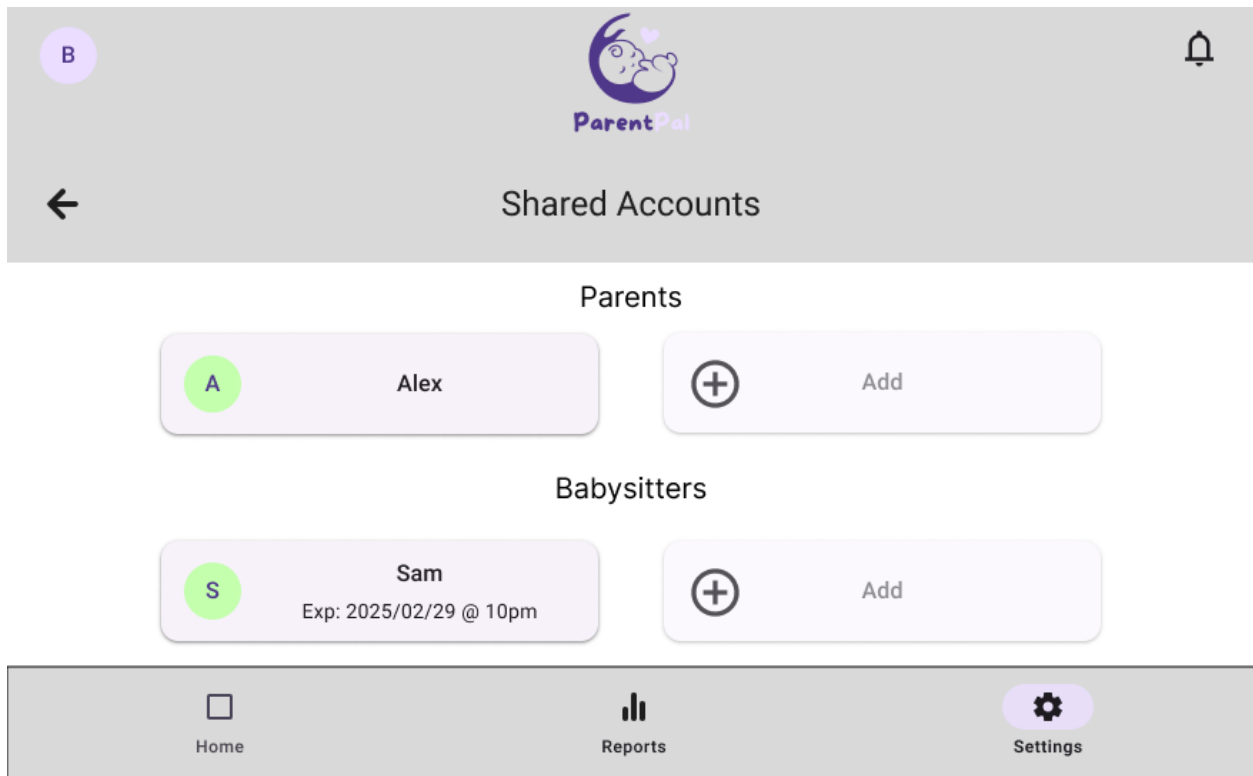
Repeat password

Save Changes

Home

Reports

Settings



←

Add Parent

Add another parent to share your little one's growth and development with.

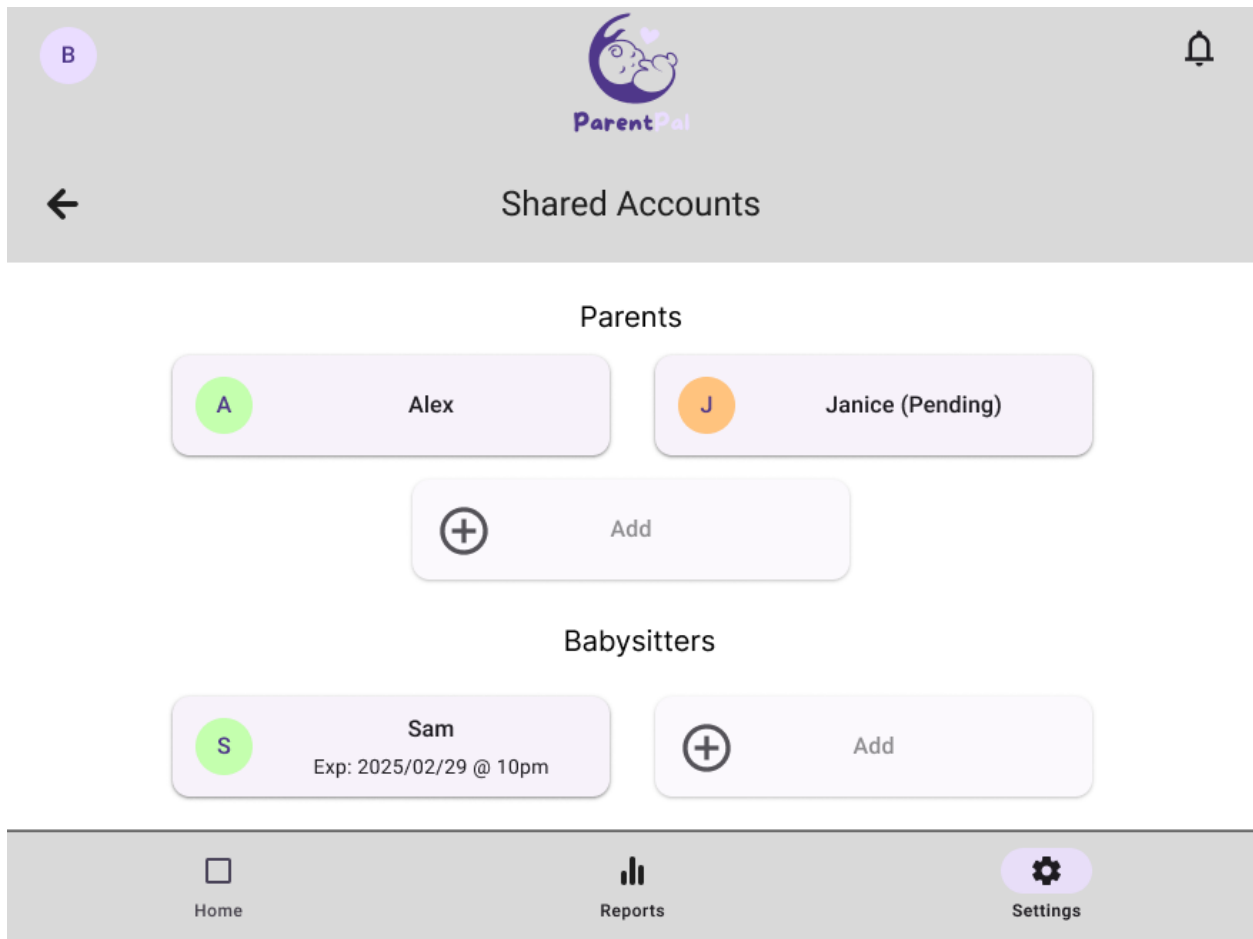
Name

Janice

Email

janice@gmail.com

Send Share Link



Add Babysitter

Add a babysitter to share your little one's schedule with and allow them to add notes, photos, and assign them tasks.

Name

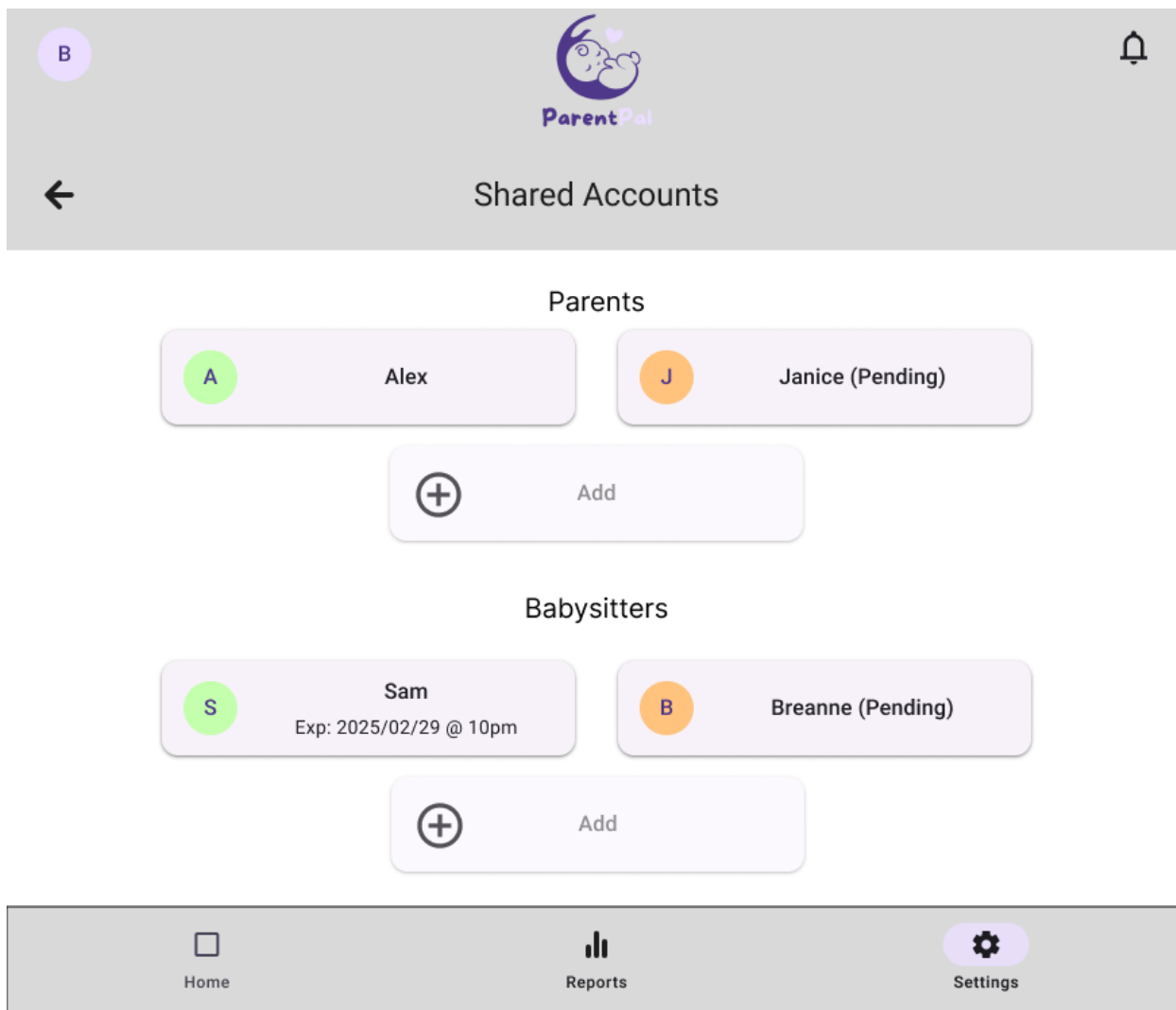
Breanne

Email

breanne@gmail.com

Add Expiry ☐

Send Share Link



Add Babysitter

Add a babysitter to share your little one's schedule with and allow them to add notes, photos, and assign them tasks.

Name

Breanne

Email

breanne@gmail.com

Start Date

2025/02/29

End Date

2025/02/29

Add Expiry



Send Share Link

Parents

A

Alex

J

Janice (Pending)



Add

Babysitters

S

Sam

Exp: 2025/02/29 @ 10pm

B

Breanne (Pending)

Exp: 2025/02/29 @ 12pm



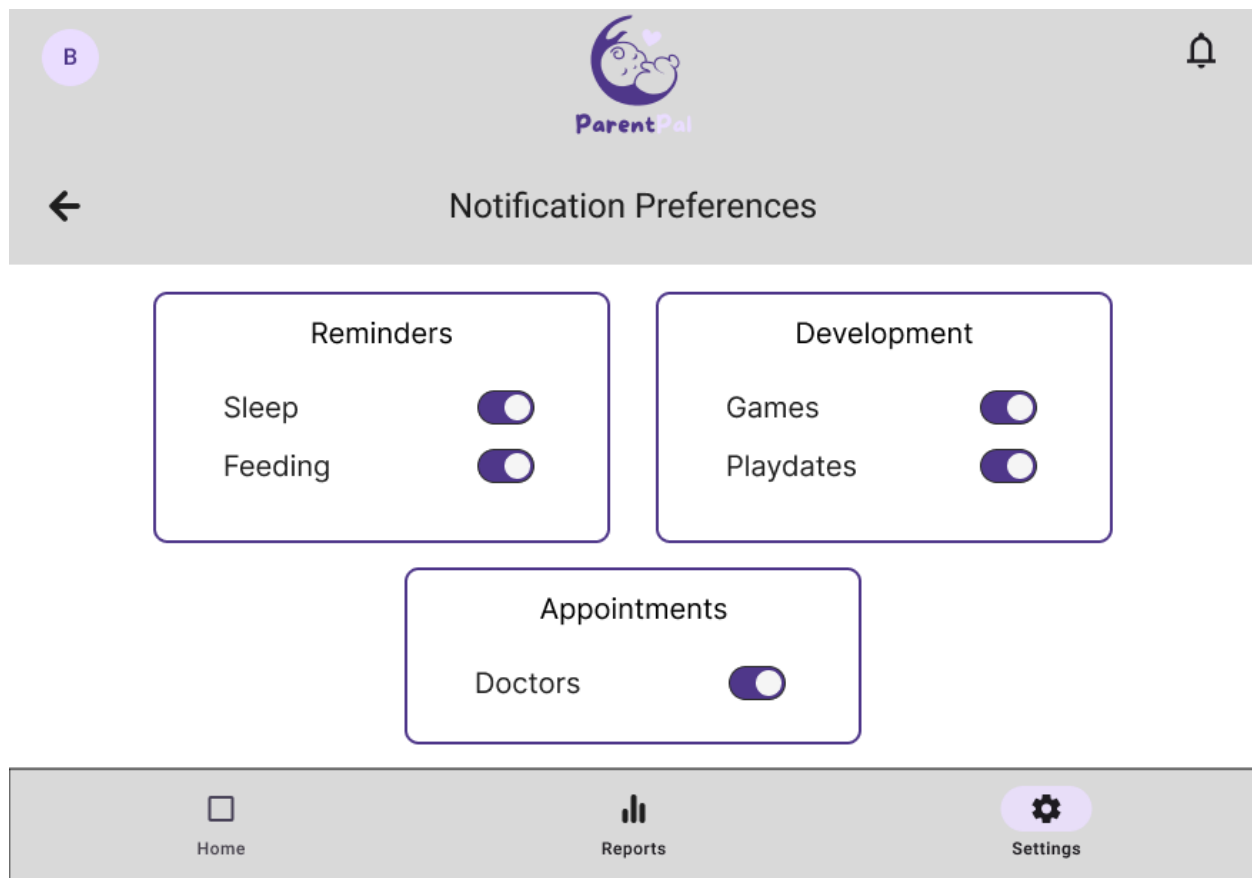
Add

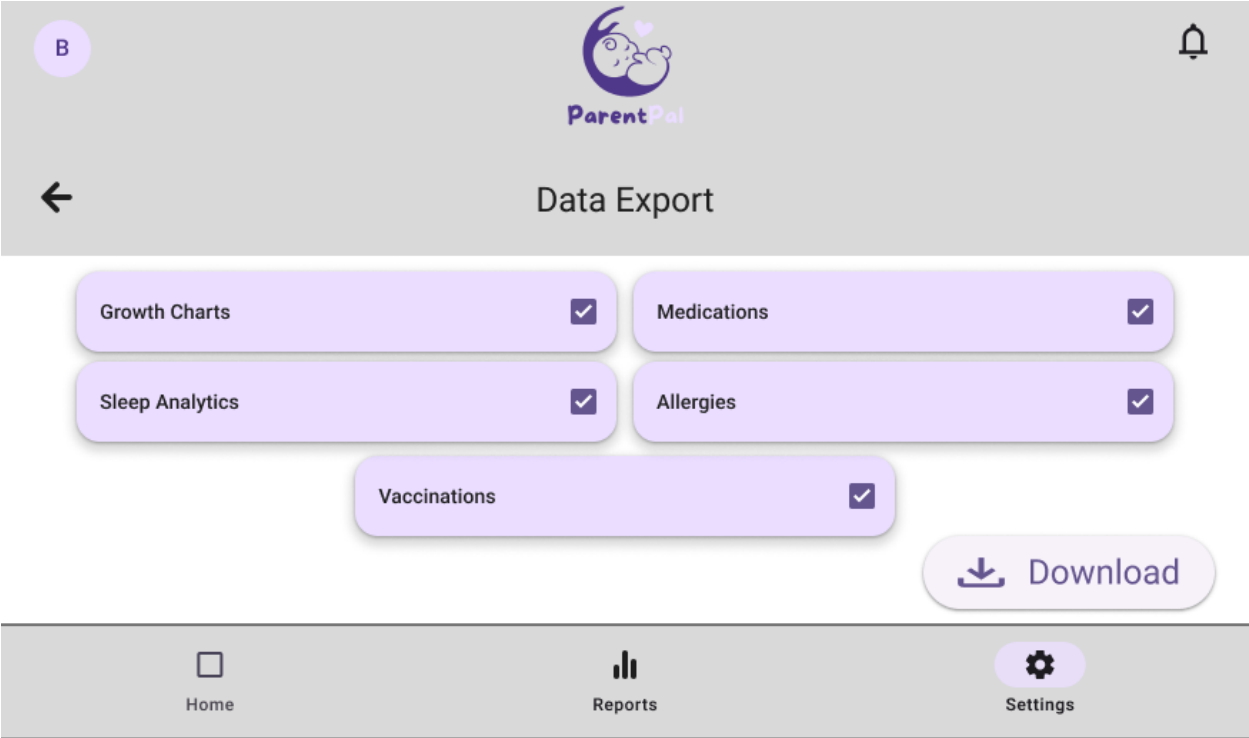
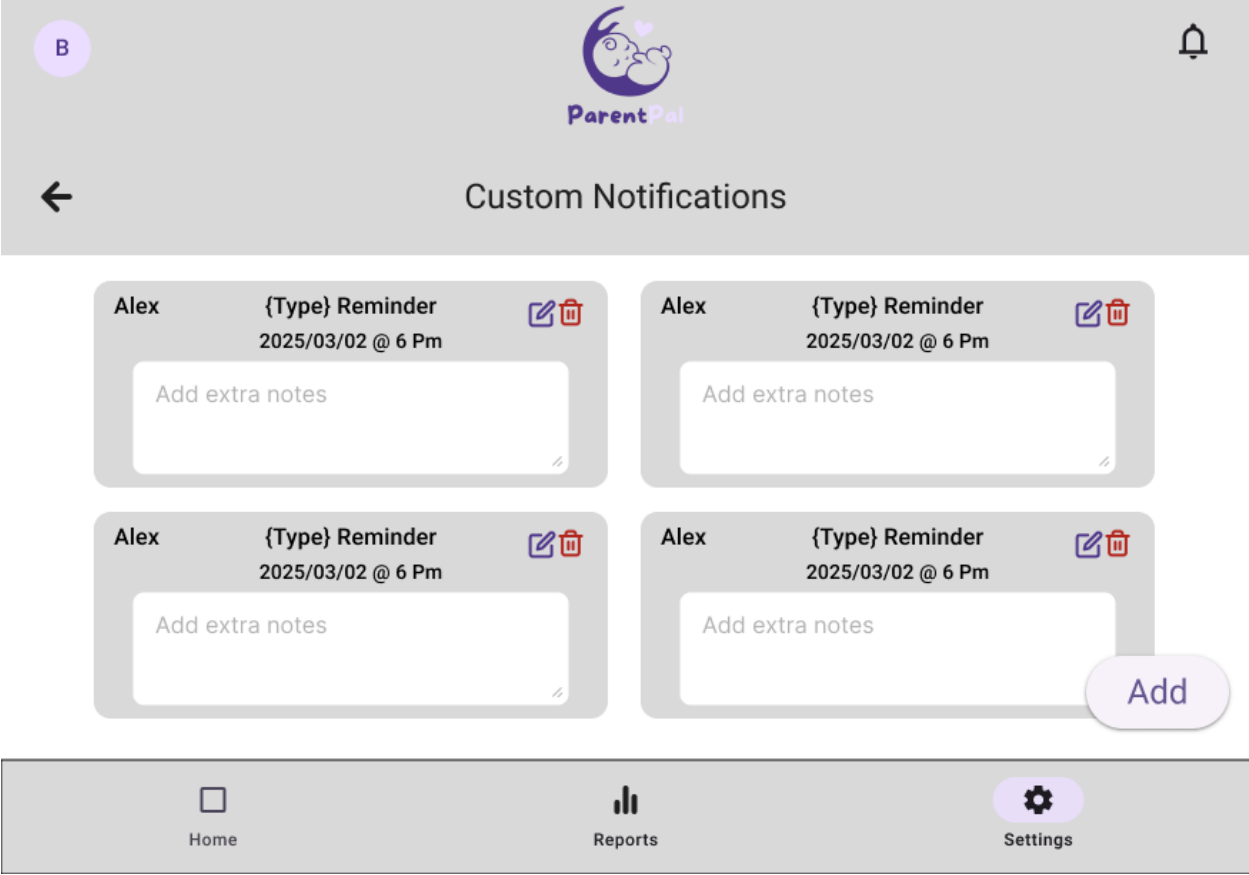
Notification Preferences

→

Custom Notifications

→





Logging - Growth Tracker, Sleep Analytics, Health Journal, Notes

B





Jack's Growth

J

Jack (< 1 Month)

2025/02/23

A

Alex (2 Years)

2023/02/23

 Photo Gallery

 Growth Tracker

 Sleep Analytics

Search for height, width, or date 



Height: 2'11" | Weight: 13 Lbs
Date: 2025/03/03

Height: 2'11" | Weight: 13 Lbs
Date: 2025/03/03

Height: 2'11" | Weight: 13 Lbs
Date: 2025/03/03

Height: 2'11" | Weight: 13 Lbs
Date: 2025/03/03

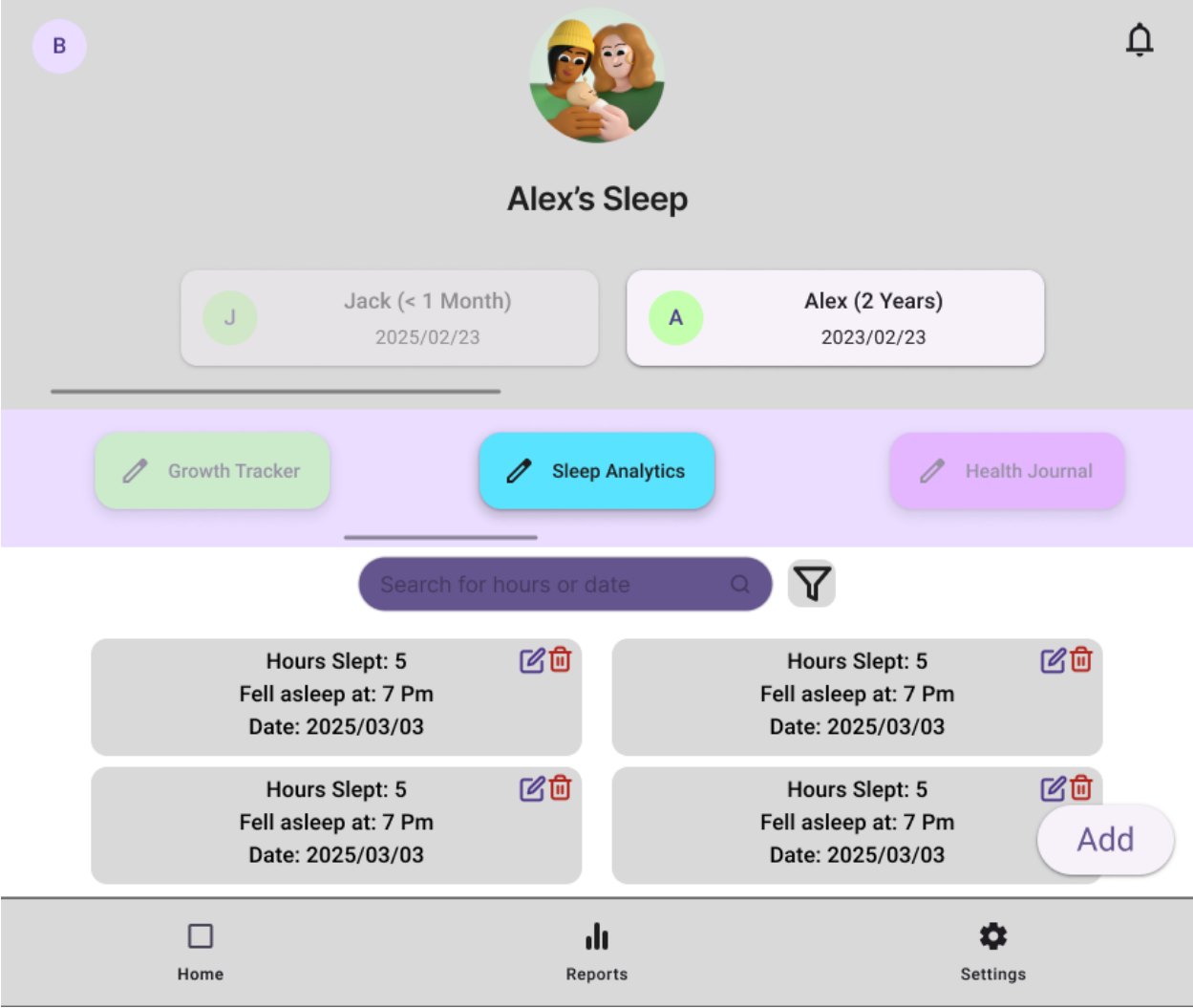
Add

 Home

 Reports

 Settings

64



B





Alex's Health Journal

J

Jack (< 1 Month)

2025/02/23

A

Alex (2 Years)

2023/02/23

 Sleep Analytics

 Health Journal

 Feeding Notes

Meds

Allergies

Vaccinations

Search for height, width, or date



Med: Tylenol

Time Taken: 6 Pm | Date: 2023/03/02

Amount: 3 Oz

Sick/Symp: Fever, pain, headache



Med: Tylenol

Time Taken: 6 Pm | Date: 2023/03/02

Amount: 3 Oz

Sick/Symp: Fever, pain, headache



Med: Tylenol

Time Taken: 6 Pm | Date: 2023/03/02

Amount: 3 Oz

Sick/Symp: Fever, pain, headache



Med: Tylenol

Time Taken: 6 Pm | Date: 2023/03/02

Amount: 3 Oz

Sick/Symp: Fever, pain, headache

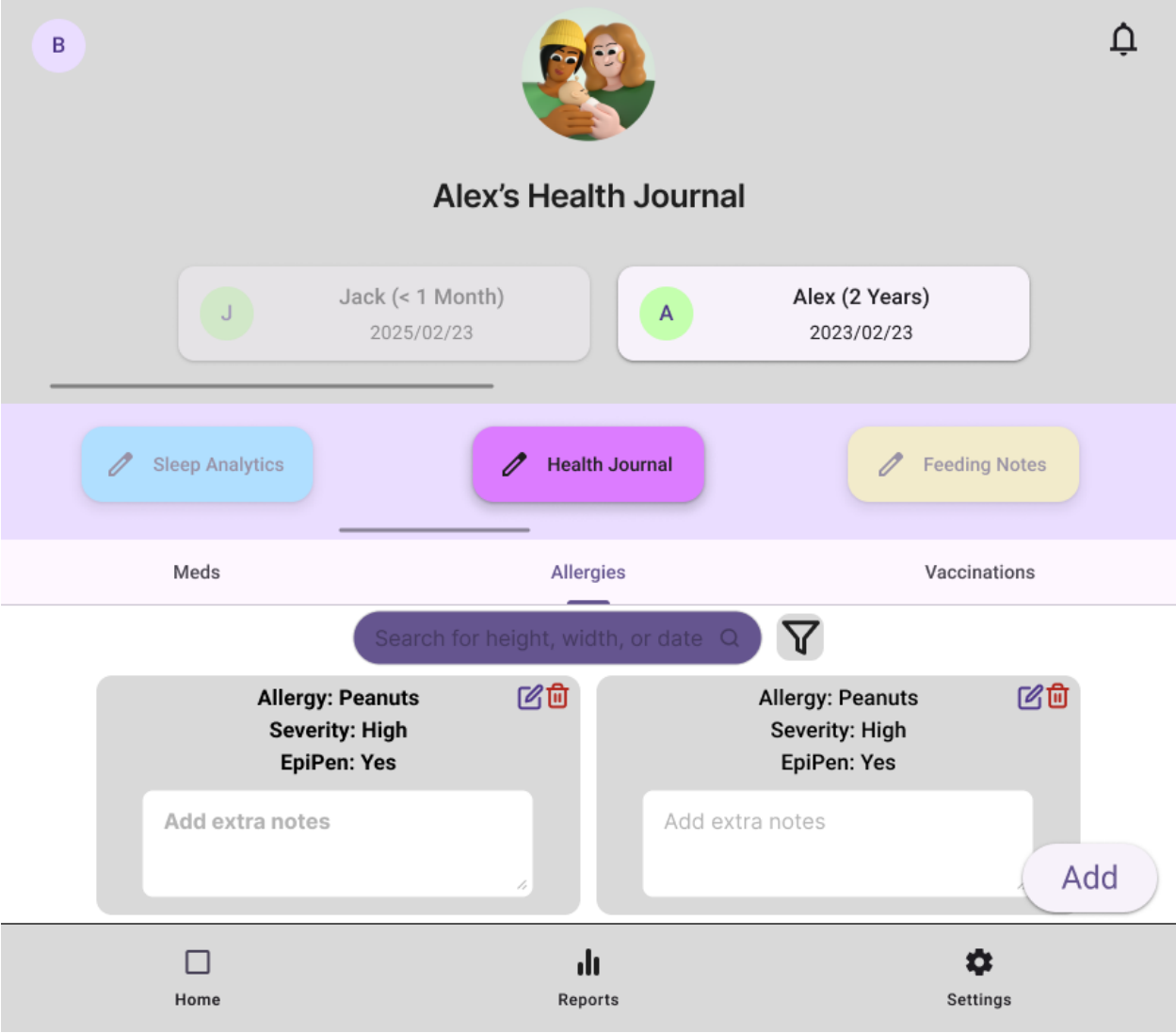


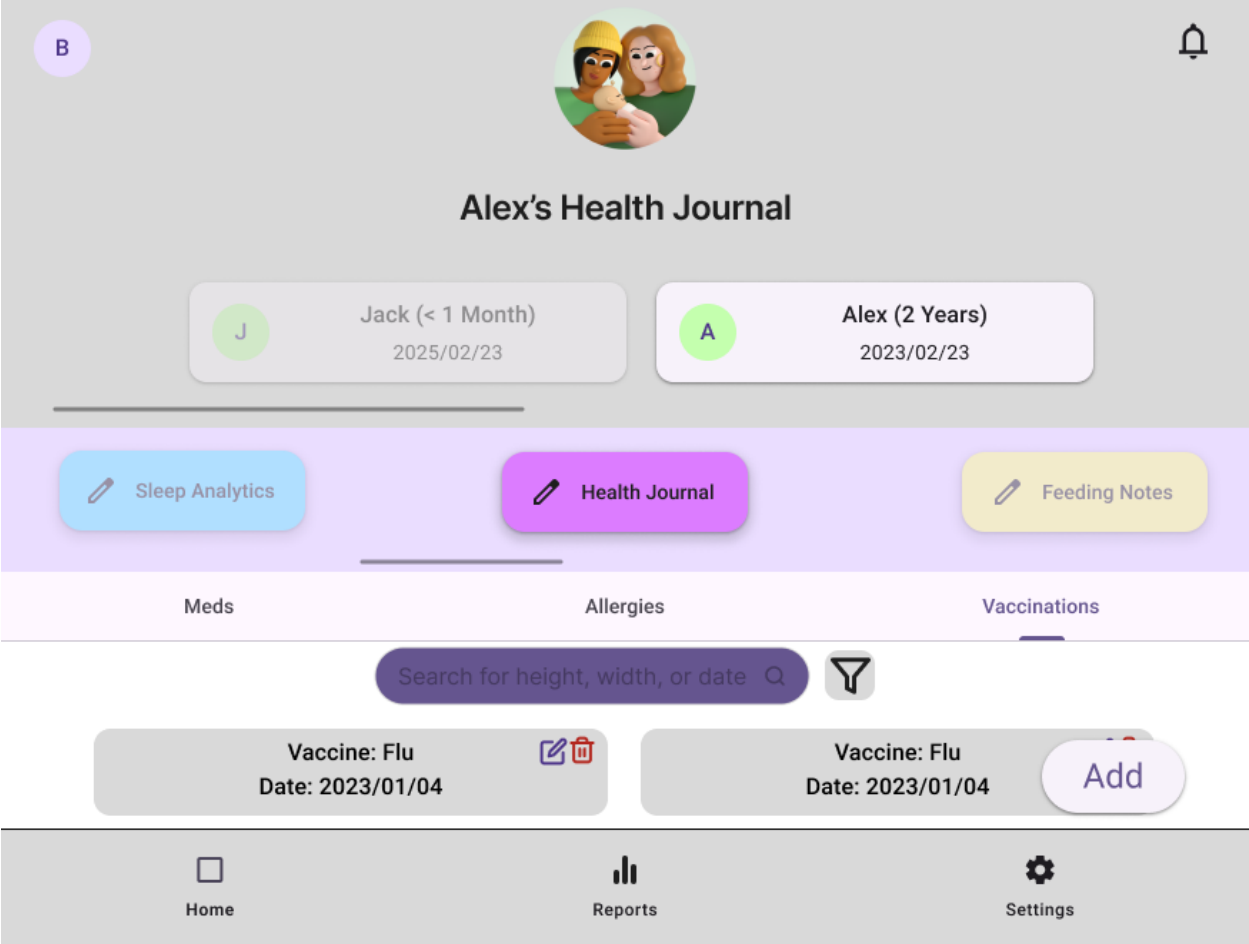
Add

 Home

 Reports

 Settings





B

Alex's Health Journal

J

Jack (< 1 Month)
2025/02/23

A

Alex (2 Years)
2023/02/23

Sleep Analytics

Health Journal

Feeding Notes

Vaccinations

Sick

Meds

Search for height, width, or date

Date: 2023/03/02
Meds Taken: Yes
Temp: 101 F

Date: 2023/03/02
Meds Taken: Yes
Temp: 101 F

Add

Home

Reports

Settings

B





Jack's Feeding Notes

J

Jack (< 1 Month)

2025/02/23

A

Alex (2 Years)

2023/02/23

 Health Journal

 Feeding Notes

 Observation Notes

Search for height, width, or date



Time: 8 am

Fed From: Bottle

Type: Milk | Amount: 5 Oz

Add extra notes



Time: 8 am

Fed From: Bottle

Type: Milk | Amount: 5 Oz

Add extra notes



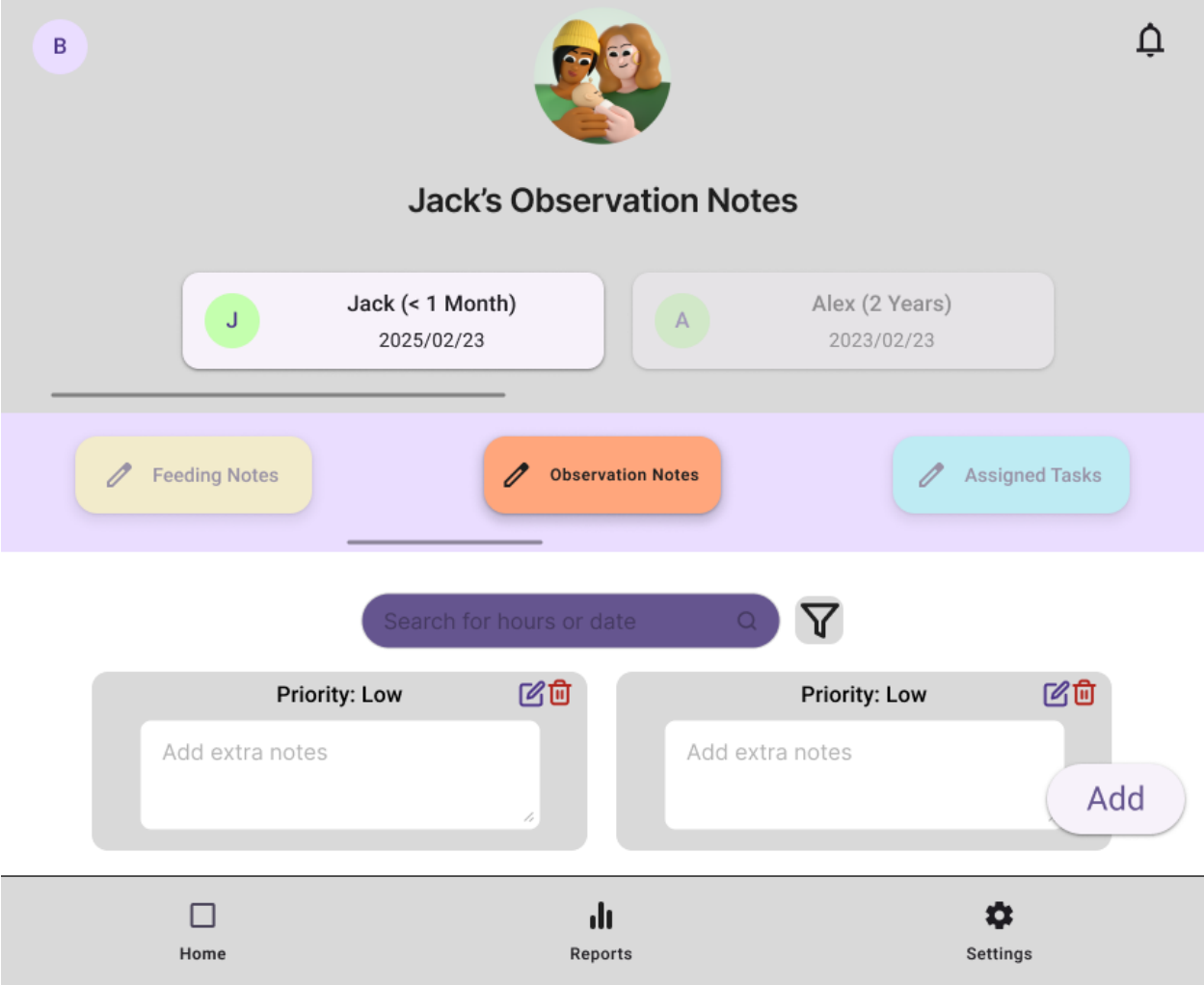
Add

 Home

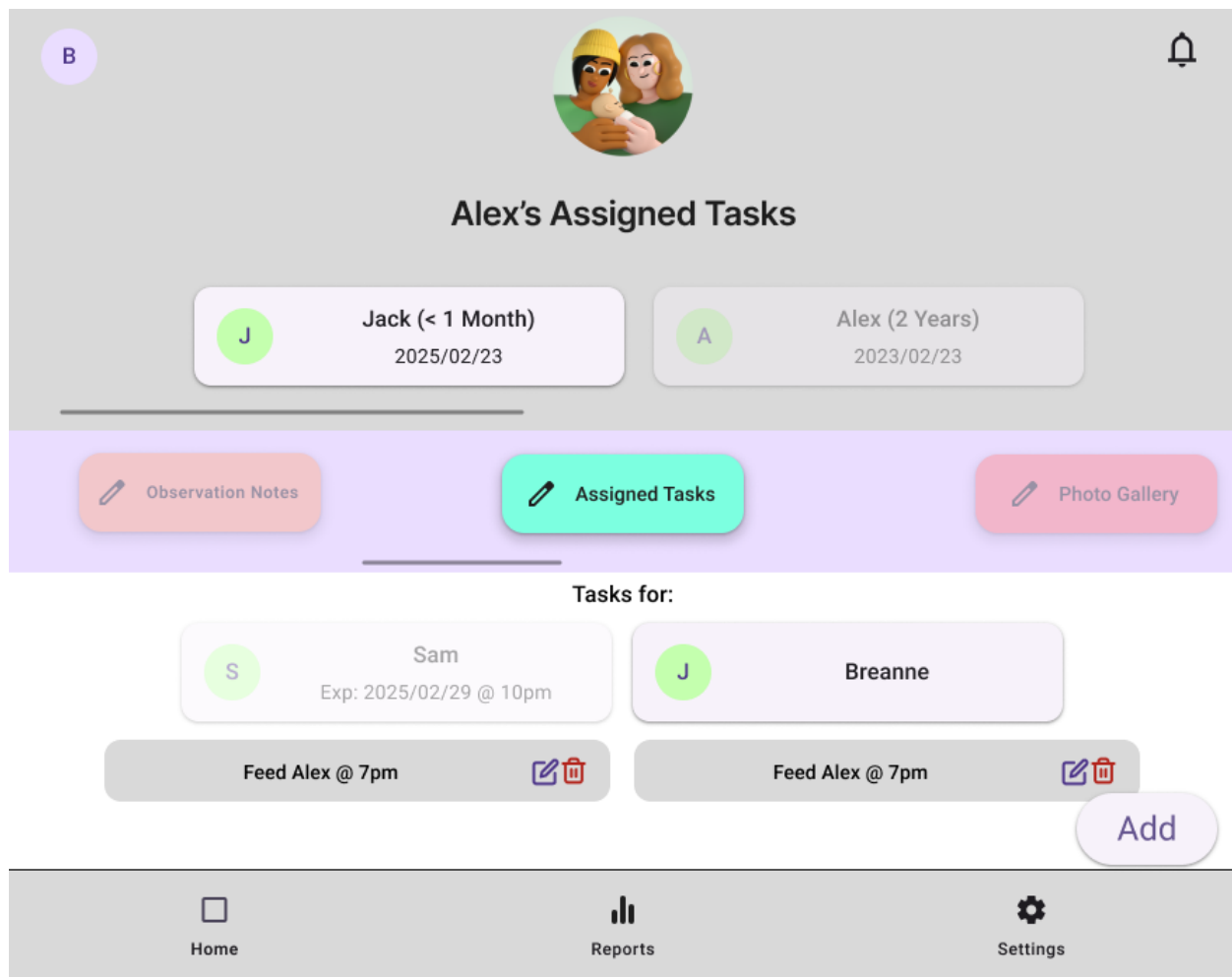
 Reports

 Settings

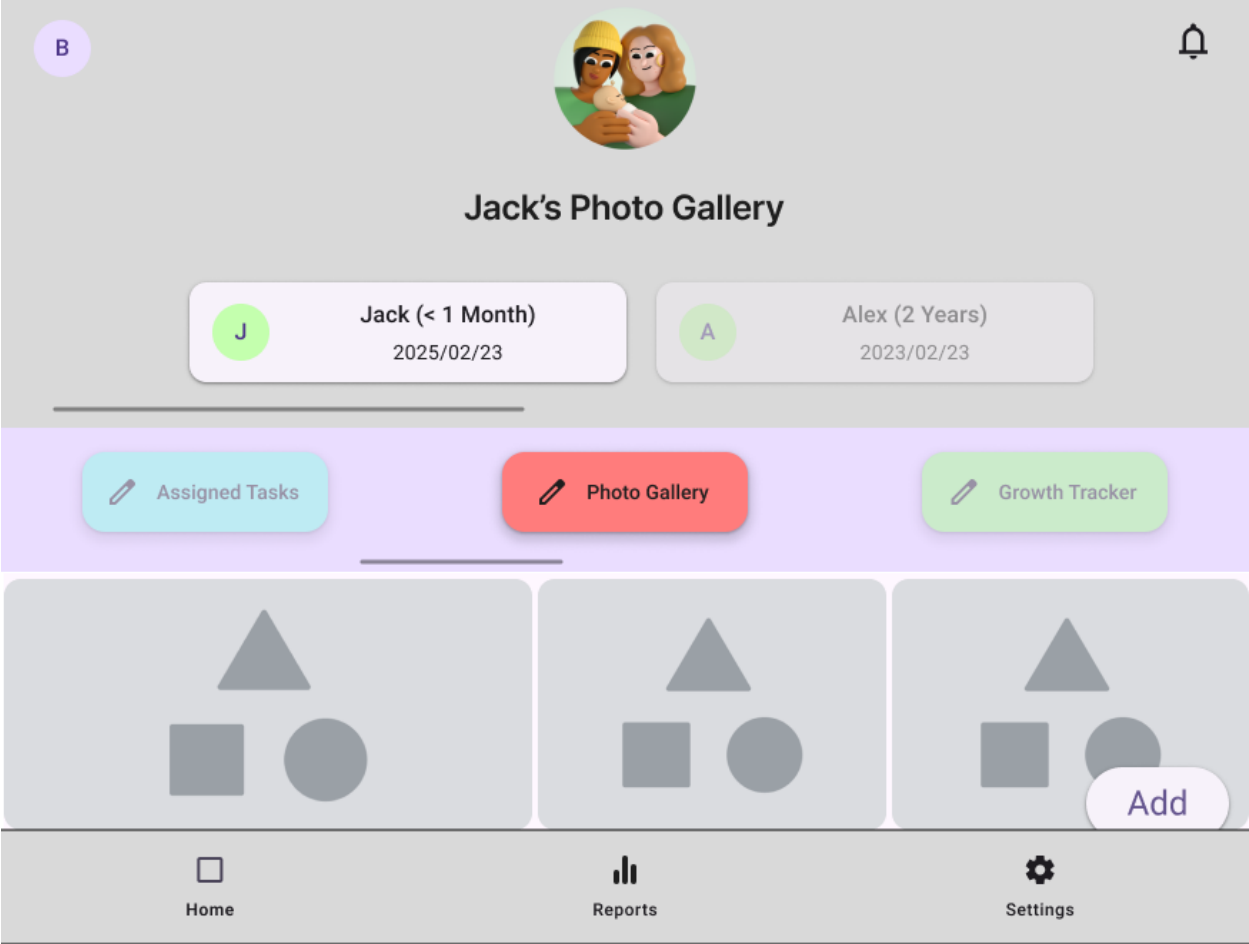
70



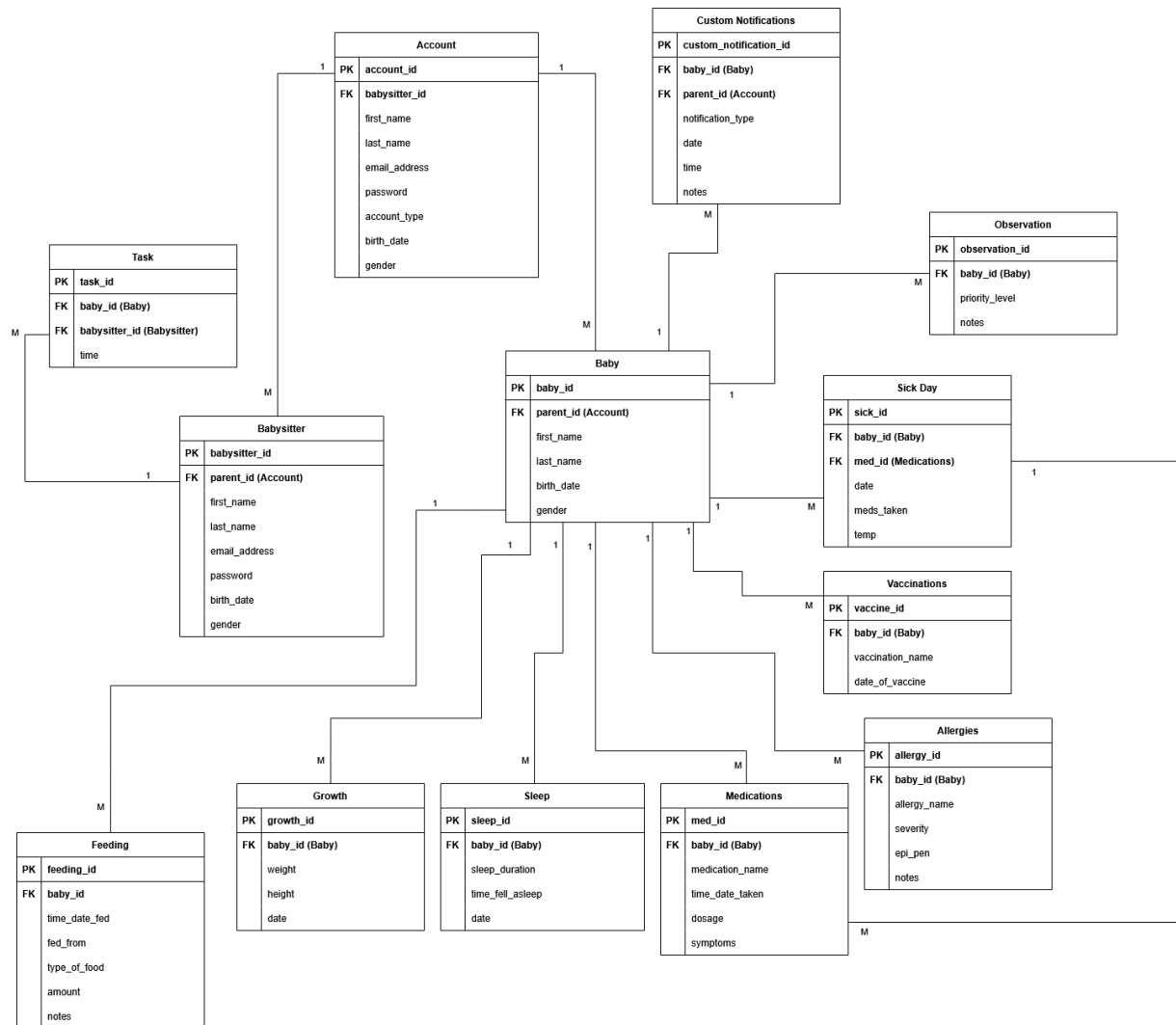
Assigned Tasks



Media Gallery



5. Database Schema & Sample Data



6. Test & Deployment Plan

6.1 Testing Strategy

The ParentPal mobile application will undergo a structured testing process to ensure its reliability, security, and usability. The following types of testing will be implemented:

- Unit Testing: Verifying individual components, such as authentication, data entry, and notification systems.
- Integration Testing: Ensuring that different modules (e.g., database, API, and UI) work together seamlessly.
- Functional Testing: Validating each feature based on expected user interactions.
- Security Testing: Identifying and mitigating vulnerabilities such as SQL injection and XSS.
- Usability Testing: Ensuring accessibility and ease of use for all users.
- Performance Testing: Assessing response times and load handling capabilities.

6.2 Testing Tools

The following tools will be used to automate and validate testing efforts:

Test Type	Tools Used
Unit Testing	Jest (JavaScript)
API Testing	Postman, Insomnia
Security Testing	OWASP ZAP, Burp Suite
Accessibility Testing	axe DevTools, WAVE, Lighthouse
UI Testing	Selenium

6.3 Functional Testing Plan

Each feature of ParentPal will be tested to verify correct behavior. Below is a sample of the test cases:

Feature	Test Cases	Expected Outcome
Authentication	Sign in with valid/invalid credentials	Successful login or appropriate error
	Sign up with valid/invalid details	User created or validation error
Growth Tracker	Log height, weight and date	Data correctly stored and displayed
	Compare with standard growth charts	Accurate results displayed
Sleep Analytics	Log sleep hours, times and dates	Data stored and insights generated
	Compare average hours slept	Accurate results displayed
Health Journal	Log medications, allergies, vaccinations, and sick days	Data recorded accurately
	Compare average sick hours	Accurate results displayed
Notes	Log observation notes and feeding notes	Data recorded accurately
Photo/Video Storage	Upload/delete media	Properly stored or removed
Data Export	Create/Delete PDF	Properly stored or removed
Shared Accounts	Add/Delete Shared Account	Properly stored or removed

Custom Notifications	Create, Edit, or Delete notifications	Properly stored, edited, or removed
----------------------	---------------------------------------	-------------------------------------

6.4 Security Testing

To ensure ParentPal is secure, the following test cases will be implemented:

Vulnerability	Test Cases	Expected Outcome
SQL Injection	Enter SQL code in input fields	Input sanitized, no database access
Cross-Site Scripting (XSS)	Inject malicious scripts	Script blocked, no execution
Session Hijacking	Attempt to use stolen session tokens	Sessions expire after inactivity

6.5 Accessibility Testing

Accessibility is a key focus to ensure usability for all users. Tests include:

- Keyboard navigation: Verify all elements are navigable.
- Color contrast: Ensure readability.
- ARIA roles and labels: Implement for assistive technology compatibility.
- Form labels and alt text: Validate for completeness.

6.6 Hosting Solution

Hosting solutions for ParentPal will be:

- Frontend Hosting: Firebase Hosting (Free tier with scalable options)
- Backend API: Render.com (Free plan for initial deployment, scales as needed)
- Database: PostgreSQL on Render.com (Free tier, upgrade as required)
- Authentication: Firebase Authentication

6.7 Deployment Process

Deployment will follow a CI/CD pipeline using GitHub Actions:

1. Development: Code commits trigger automated tests.

2. Staging: Successful tests deploy to a staging environment for manual review.
3. Production: Approved changes are pushed to the live server.

6.8 Estimated Cost (3 months)

Service	Provider	Cost Estimate (3 months)
Frontend Hosting	Firebase	Free (Pay for overages)
Backend API	Render	Free (Upgradable)
Database Storage	Render	\$7/month (Optional)
Authentication	Firebase	Free (Up to 10k users)
Domain Name	NameCheap	\$4.25
Total	-	\$0 - \$25.25

6.9 Access Management

A domain name ParentPals.ca will be used for application access (Another domain name will be used if ParentPals.ca is taken before the application is built). Firebase Hosting provides a free SSL certificate, ensuring secure communication.

6.10 Conclusion

This Test & Deployment Plan ensures that ParentPal is thoroughly tested, secure, and reliably hosted, with a clear strategy for deployment and cost management.

6.11 Development Timeline

Milestone	Start Date	Tasks Planned
0	2025-09-03	- Review proposal, finalize frameworks - Install any needed tools, IDEs, configure dev environment - Set up GitHub repository with approved proposal & framework documents
1	2025-09-08	Sign Up/Sign In (Parent/Babysitter): - Front-end & backend implementation - Firebase authentication Database (Parent/Babysitter): - Design database schema (PostgreSQL)
	2025-09-15	Growth Tracker (Parent): - Front-end & back-end implementation - Generate sleep reports Sleep Analytics (Parent/Babysitter): - Front-end & back-end implementation - Generate sleep reports Testing: - Unit testing for data validation on growth/sleep tracking
	2025-09-22	Health Journal (Parent/*Babysitter): - Front-end & back-end implementation - *Generate health reports Observation/Feeding Notes (Parent/*Babysitter): - Front-end & back-end implementation - *Generate notes reports Testing: - Unit testing for data validation on health/notes tracking Reports: - Create and submit progress report and demo video

		*Limited functionality
2	2025-09-29	Settings (Parent/*Babysitter): - Front-end & back-end implementation Deployment Prep: - Deploy back-end (Render.com) and front-end (Firebase) Shared Accounts: - Invite babysitter (Role-based access) Testing: - Security Scans (OWASP ZAP)
	2025-10-06	Extended Features: - Custom notifications (Parent) - Photo gallery (Parent/*Babysitter) - Assigned Tasks (Parent)
	2025-10-13	Break Week - Fix deployment issues - Polish UI/UX (Accessibility checks)
	2025-10-20	Data Export: - Generate PDF Reports Testing: - Integration testing (Postman) Reports: - Catch up time allocated - Create and submit progress report and demo video - Deploy initial progress to hosting platform
3	2025-10-27	Optimization: - Performance tuning (load testing) - Error handling/logging - GitHub documentation updates
	2025-11-03	Testing: - Finalize testing (Usability, accessibility) Bug Fixes & Polish: - Address feedback from demo videos - Edge-case testing (eg. offline mode)

	2025-11-10	Documentation: - User manual - API documents - Finalize github README file Reports: - Create and submit progress report and demo video
4	2025-11-17	- Last minute changes/edits
	2025-11-24	Final Report Week: - No new development - Write project report/ presentation video - Make sure I have everything needed to be done for uploading
	2025-12-01	- Create and submit final report and presentation video - Upload all needed deliverables to LMS
Link to Capstone Github Repository: https://github.com/BraedenLyman/ParentPal		
Final Hosted Project Due: December 5th, 2025 - no extensions		

7. Reflection on the Proposal Process for ParentPal

7.1 Introduction

The process of preparing a detailed proposal before diving into the actual development of my capstone project, **ParentPal**, has been an invaluable exercise in structured planning. A well-crafted proposal serves as a roadmap, ensuring that the project's vision aligns with its technical execution while minimizing wasted effort and scope creep. By meticulously outlining the app's purpose, features, user roles, database structure, testing strategy, and development timeline, I was able to identify potential challenges early and establish a clear direction before writing any code.

This reflection examines the most useful elements of the proposal, their impact on the development process, and how they will contribute to the success of **ParentPal**. Additionally, I will discuss how certain deliverables helped refine my approach and why comprehensive planning is essential for any software project.

7.2 Project Overview and Scope Definition

One of the most critical sections of the proposal was the **Project Overview**, which forced me to concisely articulate **ParentPal's** core purpose: a tool for tracking a baby's development through features like **Growth Tracker, Sleep Analytics, and Health Journal**. Without this clarity, I might have been tempted to add unnecessary features, leading to **scope creep**—a common pitfall in software development.

The **Scope Definition** section was equally important because it established **technical constraints** (e.g., no ORM frameworks, SQL injection protection) and **functional requirements** (e.g., mobile-first design, accessibility compliance). By explicitly stating these boundaries, I could avoid wasting time on incompatible technologies or tangential features that don't align with the project's primary goals.

Impact on Development

- **Prevents feature bloat** by keeping the focus on essential functionalities.
- **Reduces technical debt** by enforcing best practices (e.g., security measures) from the start.
- **Ensures consistency** in design and functionality by setting clear expectations.

7.3 Functionality Chart and User Roles

The **Functionality Chart** was one of the most practical components of the proposal. By breaking down features into a **matrix comparing Parent vs. Babysitter permissions**, I could visualize access disparities early in the process. For instance, I realized that babysitters should **not have access to growth records**, which directly influenced both the **database schema** and **UI design**.

Key Insights from User Role Mapping

- **Differentiated workflows**: Parents and babysitters require distinct interfaces and permissions.
- **Security considerations**: Role-based access control (RBAC) must be implemented to prevent unauthorized data modifications.
- **Future scalability**: The chart makes it easier to add new user roles (e.g., pediatricians) later.

Without this preemptive planning, I might have had to **refactor large portions of the codebase** later to accommodate these differences, leading to delays and frustration.

7.4 Screen Mockups

Creating **Screen Mockups** before writing any code was an absolute necessity. These mockups served as a **visual blueprint**, ensuring consistency in the app's flow and layout. For example:

The **Sign-Up Sequence** mockups revealed that parents and babysitters needed **divergent paths after registration**, which informed backend logic.

- **Desktop/Landscape View** mockups highlighted **responsive design challenges**, allowing me to address CSS issues early rather than scrambling for fixes later.

Why Mockups Are Indispensable

- **Clarifies user experience (UX)**: Helps identify confusing navigation or missing steps.
- **Facilitates stakeholder feedback**: Allows for early adjustments before development begins.
- **Reduces UI/UX rework**: Ensures alignment between design and functionality from the start.

Had I skipped this step, I might have ended up with an inconsistent or unintuitive interface, requiring significant revisions post-development.

7.5 Database Schema

Designing the **Database Schema** upfront was crucial for avoiding **mid-project restructuring chaos**. By mapping out tables (e.g., Account, Growth, etc) and relationships (e.g., one Account to many Babies), I established a solid foundation for data management.

Benefits of Early Schema Design

- **Prevents data redundancy**: Proper normalization ensures efficient storage.
- **Enhances security**: Clearly defines where access restrictions (e.g., babysitter permissions) should be enforced.
- **Ensures data integrity**: Validation rules (e.g., vaccination date checks) are implemented consistently.

If I had postponed schema design, I might have encountered **performance bottlenecks, security flaws, or data corruption issues**—problems that are much harder to fix after development is underway.

7.6 Test & Deployment Plan

Many developers treat **testing and deployment** as afterthoughts, but outlining these processes in the proposal ensured they were integrated into the workflow from the beginning.

Testing Strategy

- **Unit Testing (Jest)**: Ensures individual components work as expected.
- **Security Testing (OWASP ZAP)**: Identifies vulnerabilities like SQL injection risks early.
- **User Acceptance Testing (UAT)**: Validates that features meet real-world needs.

Deployment Considerations

- **Hosting Research (Firebase, Render.com)**: Avoids last-minute surprises like unexpected costs or scalability limitations.
- **CI/CD Pipeline**: Automates builds and deployments to reduce manual errors.

By planning for testing and deployment upfront, I can ensure a **smoother release process** and a more stable final product.

7.7 Development Timeline

A well-structured **Development Timeline** is essential for maintaining accountability and ensuring steady progress. Breaking the project into **manageable milestones** (e.g., "Sign-Up by Week 1," "Reports by Week 3") helped me:

- **Identify dependencies** (e.g., the Health Journal requires the database schema first).
- **Avoid disjointed development** (prevents coding features in an illogical order).
- **Stay on track** by setting realistic deadlines.

Without a timeline, I might have **prioritized flashy features over foundational ones**, leading to integration headaches later.

7.8 Conclusion: The Strategic Value of a Detailed Proposal

Preparing a comprehensive proposal for **ParentPal** was far more than a bureaucratic exercise—it was a **strategic investment** in the project's success. By defining the **scope, functionality, user roles, database structure, testing approach, and timeline** before writing any code, I established a **clear roadmap** that will guide development efficiently.

Key Takeaways

1. **Prevents Scope Creep:** A well-defined project overview keeps development focused.
2. **Reduces Rework:** Early mockups and schema design minimize late-stage changes.
3. **Enhances Security & Usability:** Functionality charts and testing plans ensure robustness.
4. **Maintains Accountability:** A development timeline keeps progress measurable.

Ultimately, the proposal process has **saved time, reduced errors, and set ParentPal on a path toward a polished and functional final product**. Moving forward, this document will serve as a **living reference**, helping me stay aligned with the original vision while adapting to real-world challenges.

For future projects, I will **always prioritize thorough planning**—because in software development, **the time spent refining the blueprint is just as critical as the time spent building**.

7.9 A Note to My Future Self:

As you dive into the development of **ParentPal** and prepare to graduate, remember this: **Stay disciplined, but stay flexible**. You've laid a strong foundation with your proposal—trust the process, but don't be afraid to adapt when challenges arise. Keep breaking tasks into small, manageable steps, and hold yourself accountable to the timeline you've set. When frustration or self-doubt creeps in (because it will), remind yourself why you started: to build something meaningful that solves real problems. Don't cut corners on testing, documentation, or user feedback—these are what separate a functional project from a polished, professional one. Most importantly, celebrate the small wins along the way. You're capable of far more than you think, but progress happens one commit at a time. When you look back on this, let it be with pride not just in the final product, but in the growth it took to get there. **Keep going**.